

Forecast of solar cycle 25 based on Hybrid CNN-Bidirectional-GRU (CNN-BiGRU) model and Novel Gradient Residual Correction (GRC) technique

Abhijeet Kumar, Vipin Kumar*

Mahatma Gandhi Central University, Motihari, Bihar 845401, India

Received 21 August 2023; received in revised form 26 December 2023; accepted 9 January 2024

Available online 11 January 2024

Abstract

In this study, a hybrid deep-learning(DL) model is proposed, which consists of a Convolution-Neural-Network (CNN) and Bidirectional-Gated-Recurrent-Unit (Bi-GRU) for the prediction of sunspot numbers(SSN) of different frequencies along with novel post-processing techniques of Gradient-Residual-Correction (GRC) to enhance the accuracy of the predictions further. GRC is utilized to reduce the residual present in the forecasted values. The AdaBoost regression model is implemented over the residual obtained from the prediction of training data using a hybrid CNN-BiGRU model with respect to the gradient of the training data points to predict the residual for the test data points. Ultimately, the predicted residual for test data points is summed up with the predicted test data points to achieve the final predictions. The results obtained from the proposed methods are compared with the results obtained from traditional DL models. The validation of the proposed method is carried out based on four performance metrics, namely Root Mean Squared Error (RMSE), Mean Absolute Scaled Error(MASE), Mean Absolute Error(MAE), and Mean Absolute Percentage Error(MAPE). A significant percentage of improvement is observed while using the GRC technique in comparison to all other experimented models for all four variants of SSN data. A maximum improvement of 85.71% has been achieved in comparison to the BiLSTM model over the 13-month smoothed SSN dataset on the basis of MAPE. Friedman Ranking is also performed as a non-parametric statistical test over the results of the performance measures. This model has been utilized for the forecast of solar cycle 25(SC25) over the annual mean of total SSN. It has been observed that the SC25 is expected to reach its peak in the year 2024 with an annual average peak value of 143.641. Comparative analysis of SC25 and the peak of SSN in the ongoing cycle with the previous works are also carried out.

© 2024 COSPAR. Published by Elsevier B.V. All rights reserved.

Keywords: Sunspots; Solar cycle; Hybrid model; Model residual; Regression; Adaboost; SSN; Deep learning

1. Introduction

The appearance of sunspots on the solar surface is a major phenomenon that has a significant impact on scientific developments related to space and astronomy. The total number of sunspots that appear on the solar surface forms a pattern. This is known as the sunspot number

(SSN). This number is observed on a daily basis and recorded on a daily, monthly, and yearly basis. SSN corresponds to the solar cycle of approximately 11 years of time span. The current solar cycle is Solar Cycle 25 (SC25). Many researchers have tried to improve the accuracy of the prediction of upcoming and ongoing solar cycles. With the advent of the telescope, astronomers attempted to observe the solar surface and the appearance of sunspots (Vokhmyanin et al., 2020). Spacecraft designers consider the variances caused by sunspots as one of the significant

* Corresponding author.

E-mail address: rt.vipink@gmail.com (V. Kumar).

criteria (Marques et al., 2023). Deep-learning has proved to be a technique that helped in a significant reduction of error in the predictions of exact SSN as per the study performed by Dang et al. (2022). In their study, they compared different statistical and DL models with each other along with their combinations and found out that the ensemble done by XgBoost over the DL has the best performance based on RMSE and MAE as the performance metric. For this purpose, they utilized the monthly mean of total SSN as the dataset. They also predicted the peak for SC25 and SC26 to be 133.47 in May 2025 and 164.62 in November 2035, respectively.

Lee (2020) has applied an algorithm for frequency extraction that uses Empirical Mode Decomposition (EMD) along with LSTM to perform the predictions over the monthly averaged value of SSN and evaluated the framework over RMSE and R^2 . In this work, we have proposed a hybrid DL model composed of one-dimensional CNN followed by Bi-GRU to forecast the SSN for the ongoing solar cycle. Furthermore, a framework of prediction is proposed for the improvement of the predicted outcomes of the aforesaid DL model. In one study by Elgamal (2020), yearly mean data of SSN are used for the forecast of SC25 based on the deep LSTM model, and the model's validation is performed using MAPE and RMSE. In another study conducted by Arfianti et al. (2021), SSN data of monthly frequency are used for the comparative analysis between LSTM and GRU based on different hyperparameters. From the study of Arfianti et al. (2021), it can be deduced that GRU outperforms the LSTM model, where the two models are evaluated over MAPE. Ramadevi and Bingi (2022) proposed a non-linear autoregressive (NAR) neural network and LSTM model separately for the prediction of 13-month smoothed SSN and evaluated the results over the MSE and R^2 measures, concluding that the NAR model is achieving better performance than the LSTM model. Apart from these, Pala and Atici (2019) have also performed a forecast for SC25 explaining that the peak of SC25 is in July 2022 with 167.3 and the span of the ongoing cycle is of ten years. The actual SSN in July 2022 was 91.4, and it is following an increasing trend with a peak value of 160.5 in June 2023. For their study, they utilized the 6-part and 12-part LSTM approach and evaluated the model over monthly averaged SSN data with RMSE as the performance metric. Panigrahi et al. (2021) developed a hybrid model of ARIMA, ETS, and SVM for the SSN data of monthly and annual frequency and compared the performance with 16 different forecasting techniques, including statistical and DL approaches using MASE, RMSE and MAE measures. They predicted the time series using the hybrid model, predicted the residual using SVM, and combined the results to obtain the final forecast values.

Prasad et al. (2023) has proposed a single-layer LSTM model for the prediction of 13-month smooth SSN data and evaluated the proposed model over different sets of

intervals of the dataset with RMSE and MAE as the performance metric deducing that peak of SC25 will be achieved by April 2023 with a maximum value of 136.9. However, the actual value of SSN in April 2023 was 97.6. They have also compared the value and time of the peak of SC25 with the work of other researchers. A hybrid model of ARIMA and ANN, along with the concept of EMD, is proposed by Büyükşahin and Ertekin (2019) for the SSN forecast. They have implemented the model over the annual average SSN and evaluated it with the help of MASE, MSE and MAE. A comparative study of ARIMA, ARMA, and autoregressive Neural Networks is performed by Chattopadhyay et al. (2011) on the annual average SSN, where it is deduced that the ARNN model outperforms the other two models, claiming the better performance of neural network over the statistical methods. R^2 is used as the determining performance metric in this study. On the monthly average SSN data, a hybrid model composed of DNN and LSTM is proposed by Hasoon and AL-Hashimi (2022), which is evaluated over MSE and MAE. In this study, it was found that the model effectively captures the pattern of the SSN data.

Nghiem et al. (2022) performed SSN predictions over monthly averaged data by implementing Bayesian inference in the CNN-LSTM model and evaluated it over RMSE and MAE. They suggested that the proposed model is performing better than SARIMA, Prophet, Transformer, Informer, LSTM, and GRU. Zhu et al. (2023) predicted the SC25 based on sunspot area data using optimized LSTM, evaluating the performance of the model based on the error obtained on the peak amplitude of the last four complete solar cycles. Prasad et al. (2022) predicted the SC25 using stacked LSTM over 13-month smoothed SSN data after evaluating the performance of the model over different individual solar cycles and utilizing it as the evaluation metric. Kumar et al. (2023) proposed a novel model composed of β SARMA and LSTM and utilized it over the annual average SSN data. They evaluated the model using MdAE, SMAPE and MASE. They also compared the performance of the proposed model with different models, namely ARIMA, LSTM, MLP, and ARIMA-ANN, for three different horizons. The summarized study of the literature has been tabulated in Table 1.

Despite the implementation of the different DL and Non-DL techniques over SSN data, it is observed that some significant amount of error is introduced in the prediction due to the untreated residual rising as a result of delayed learning of the model and its inability to adapt to the distinctive crest and trough patterns in the SSN graph. To overcome this scenario, an optimized hybrid DL model (Proposed Model 1) is developed, which is able to improve the prediction results with respect to the traditional DL models. Furthermore, a novel technique (Proposed Model 2) is proposed, which is capable of improving the prediction results using the residual and gradient of the training data and its corresponding prediction.

Table 1
Literature review of the SSN prediction based upon different approaches.

Reference	Approach	Duration	Samples	Frequency
Chattopadhyay et al. (2011)	ARNN	1749–2007	258	Annual
Pala and Atici (2019)	LSTM		3240	Monthly
Büyüksahin and Ertekin (2019)	ARIMA + ANN	1700–1987	288	Annual
Elgamal (2020)	Stacked LSTM	1700–2018	318	Annual
Lee (2020)	EMD + LSTM	01/1749–10/2019	3250	Monthly
Panigrahi et al. (2021)	ARIMA + ETS + SVM	01/1749–12/2019	3252/	Monthly and
		1700–2019	320	Annual
Arfianti et al. (2021)	LSTM, GRU	01/1750–12/2019	3240	Monthly
Prasad et al. (2022)	LSTM	01/1750–12/2019	3240	Monthly
Hasoon and AL-Hashimi (2022)	DNN, RNN, DNN + LSTM	1749–2021	3266	Monthly
Dang et al. (2022)	SARIMA, Exponential Smoothing, Prophet, LSTM, GRU, Transformer, Informer	01/1749–01/2022	3277	Monthly
Ramadevi and Bingi (2022)	NAR, LSTM	1750–2020	3252	Monthly
Nghiem et al. (2022)	CNN + LSTM	01/1749–02/2022	3278	Monthly
Kumar et al. (2023)	β SARMA + LSTM	1700–2008	309	Annual

The rest of this paper is composed of a description and analysis of the dataset in Section 2, methodology in Section 3, analysis and discussion of the result in Section 4, and conclusion in Section 5.

2. Dataset

The four dataset variants, each based on a different frequency, are obtained from SILSO at the Royal Observatory in Belgium. (SILSO World Data Center, YYYY). Two of the datasets are of monthly frequency, while the other two are of yearly and daily cadence. Data with monthly cadence are based on the monthly mean and 13-month smoothing technique. The additive seasonal decomposition plots of all four variants of the utilized dataset have been illustrated in Fig. 1.

2.1. Data decomposition analysis

The seasonal decomposition depicts the three individual components of all four variants of the data for a span of twenty years for 13-month smoothed and monthly average SSN, which range from 2002 to 2022. One complete cycle is visible in the trend from 2009 to 2020, illustrating the 11-year solar cycle in Fig. 1a and 1b. Seasonality is present every year, and residual does not follow a pattern except for the increase during the peak of the cycle. Fig. 1c is the seasonal decomposition for daily data ranging from February 2022 to February 2023 with residual that follows no pattern and approximately four seasonal repetitions within a month. Fig. 1d illustrates no significant residual and seasonality for a tenure of sixty years ranging from 1960 to 2020. It follows a cyclic trend for six times in sixty years, which is evidenced by six solar cycles during this tenure.

2.2. Descriptive statistics of data

The description of the data has been tabulated in Table 2, which illustrates the time span, frequency and total number of samples in each dataset along with the range of each dataset in the form of minimum and maximum values. The -1 value present in the minimum is the indicator of the missing value; therefore, it can be discarded since the missing value is replaced in the process of data preprocessing. It can be observed from Table 2 that daily SSN has the highest number of samples, which is 75049, while an annual average of total SSN has the lowest number of samples, which is 323. The mean of all four datasets ranges between 78.379 to 81.633. The maximum value of Daily Total SSN reaches up to 528, which is an indicator of the possible existence of an outlier as it is much higher than 75% of the data, which is 123. It signifies that during the solar maximum, SSN can abruptly increase and may exist for a shorter time span. Furthermore, the range of data represents that the yearly average of total SSN has data for the longest time span starting from the year 1700 while daily SSN has been recorded after 118 years of the yearly average of total SSN beginning from the year 1818.

3. Methodology

The framework has four main components, where Pre-processing and Data Sampling is the first component, followed by the second component of Model Training and Optimization, which includes the proposed hybrid DL model CNN-BiGRU (Proposed Model 1) as depicted in Fig. 2 and its hyperparameter tuning. The third component is the post-processing technique of GRC (Proposed Model 2) for handling the error produced by the residual of the prediction, which is illustrated in Fig. 4. The proposed

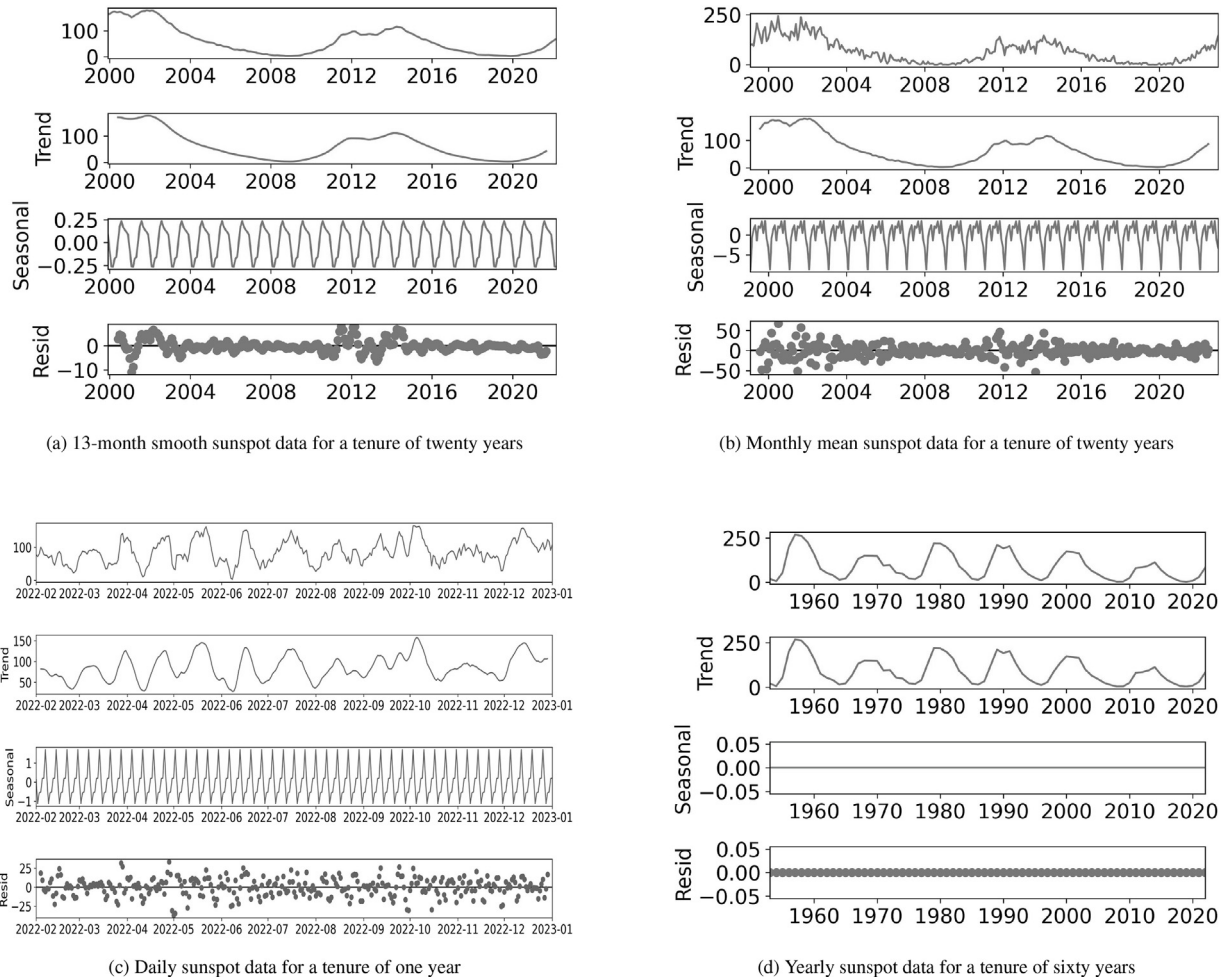


Fig. 1. Additive Decomposition of sunspot data with different frequencies.

Table 2

Descriptions of four variants of sunspot data (SILSO World Data Center, YYYY).

Name of Data	Time Span	Frequency	Samples	Min	25%	50%	75%	Max	Mean
Daily Total SSN	8/1/1818–30/06/2023	Daily	75049	−1	15	58	123	528	78.748
Monthly Mean Total SSN	1/1749–1/2023	Monthly	3289	0	24	67.2	122.4	398.2	81.633
13-month Smoothed Monthly Total SSN	7/1749–12/2022	Monthly	3282	−1	25	71.6	118.8	285	81.559
Yearly Mean Total SSN	1700–2022	Yearly	323	0	24.45	65.3	114.95	269.3	78.379

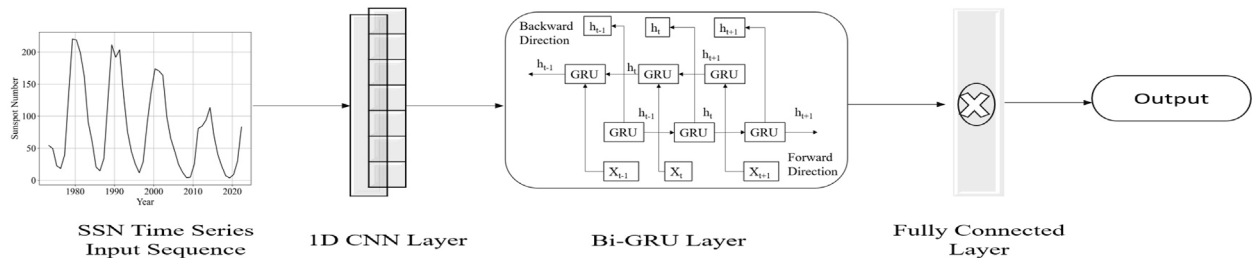


Fig. 2. CNN-BiGRU (Proposed 1) architecture for the prediction of SSN.

models are compared with traditional DL models, namely BiLSTM, LSTM, GRU, CNN, and RNN. For this, the models are evaluated on different parameters, which

include RMSE, MASE, MAE, and MAPE. Hence, the last component of the framework is the model evaluation component.

3.1. Input data

SSN feature is pulled out from the data files obtained from the SILSO World Data Center. Let $x_t \in R$ be the observed time series element. The univariate dataset can be written as in the Section 2,

$$\exists D_T = \{x_t | x_t \in R, t \in T\} \quad (1)$$

where $T \subseteq R, t = 1, 2, 3, \dots, T$. It is assumed that the data does not have high variability, lack of stability or absorption state.

3.2. Data preprocessing

The univariate data treated for missing values, which are represented as -1 in the dataset. The missing values x_t at time t are replaced with the mean of one datapoint, which is immediately before and after the missing value as depicted in Eq. 2.

$$x_t = \frac{(x_{t+1} + x_{t-1})}{2}, \forall x_t \in D_T \quad (2)$$

where x_t represents the missing data point and x_{t+1} and x_{t-1} represent the data point immediately ahead and before the missing value. Then, the z-score normalization is carried out on the dataset using the formula given in Eq. 3, where μ represents the mean, σ represents the standard deviation, and x'_i represents the standardized data point.

$$x'_i = \frac{(x_i - \mu)}{\sigma}, \forall x_i \in D_T \quad (3)$$

where standardized data $D'_T = \{x'_i\}_{i=1}^n$, mean $\mu = \frac{1}{n} \sum_{i=1}^n (x_i)$, T is the total time period, and standard

deviation $\sigma = \left(\sqrt{\frac{\sum_{i=1}^n (x_i - \mu)^2}{n}} \right)$. Then, the data is divided

into a 90:10 ratio of training and test data, respectively. Out of the training data, 10% of the data is pulled out for validation purposes. Therefore, the dataset is finally divided in the ratio of 81:9:10. The resultant dataset will be $D'_{Tr} \subseteq D'_T, D'_{Va} \subseteq D'_T$ and $D'_{Te} \subseteq D'_T$. Furthermore, the three resultant datasets are again divided into x component and y component individually resulting in $x_{tr}, x_{va}, x_{te}, y_{tr}, y_{va}$ and y_{te} .

3.3. The proposed methods

Two methods are proposed: CNN-BiGRU (Proposed Model 1) for the effective prediction of SSN and GRC (Proposed Model 2) for improving the prediction results obtained from the former model. The first model is a hybrid DL model with optimized parameters, while the latter method is a post-processing technique that works in association with the first method to improve the prediction results by correcting the error produced by the residual of the predictions.

3.3.1. Proposed Model 1: Convolutional Neural Network-Bidirectional Gated Recurrent Unit (CNN-BiGRU)

The proposed model is composed of a single layer of one-dimensional CNN with 12 filters of kernel size 12 with a 'relu' activation function followed by a Bidirectional GRU with 16 units and a 'relu' activation function. Finally, a fully connected layer of one unit is attached for the one-step ahead forecast. Model is trained with an optimized batch size of 66 and an 'Adam' optimizer. The model is trained for 250 epochs with enabled early stopping, which monitors validation loss with a patience value of 20. The loss is calculated as a mean squared error. The model is validated based on the plots of validation loss catching with the training loss following a gradual decreasing pattern and tending to a minimum constant value. A schematic diagram of the proposed model is illustrated in Fig. 2. The dimensions in every step are represented in Fig. 3.

- Convolutional Neural Network (CNN-1D) Layer

$$h_i = f \left(\sum_{j=1}^m w_j x_{i+j-1} + b \right) \quad (4)$$

In the Eq. 4, h_i represents the output at the i^{th} position, and f is the representation of the activation function with w_j denoting the weight of j^{th} filter. x_{i+j-1} is the representation of the value at position $i + j - 1$ and b is the bias.

- BiDirectional Gated Recurrent Unit

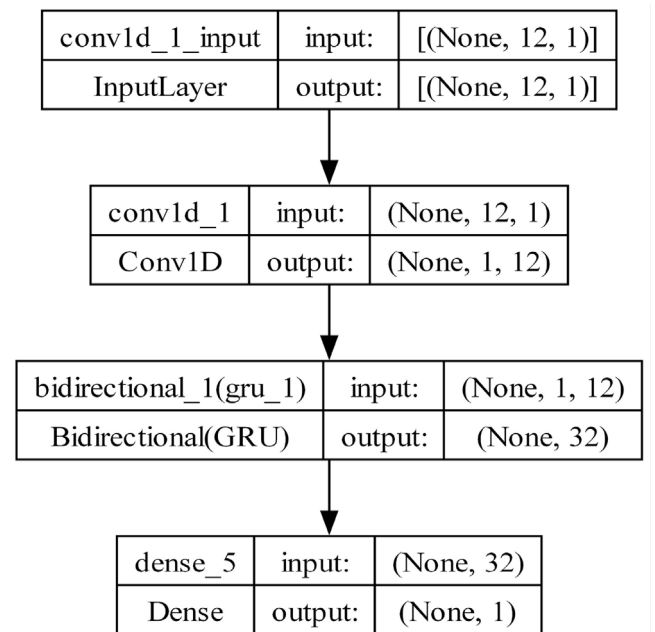


Fig. 3. CNN-BiGRU (Proposed Model 1) architecture for the prediction of SSN.

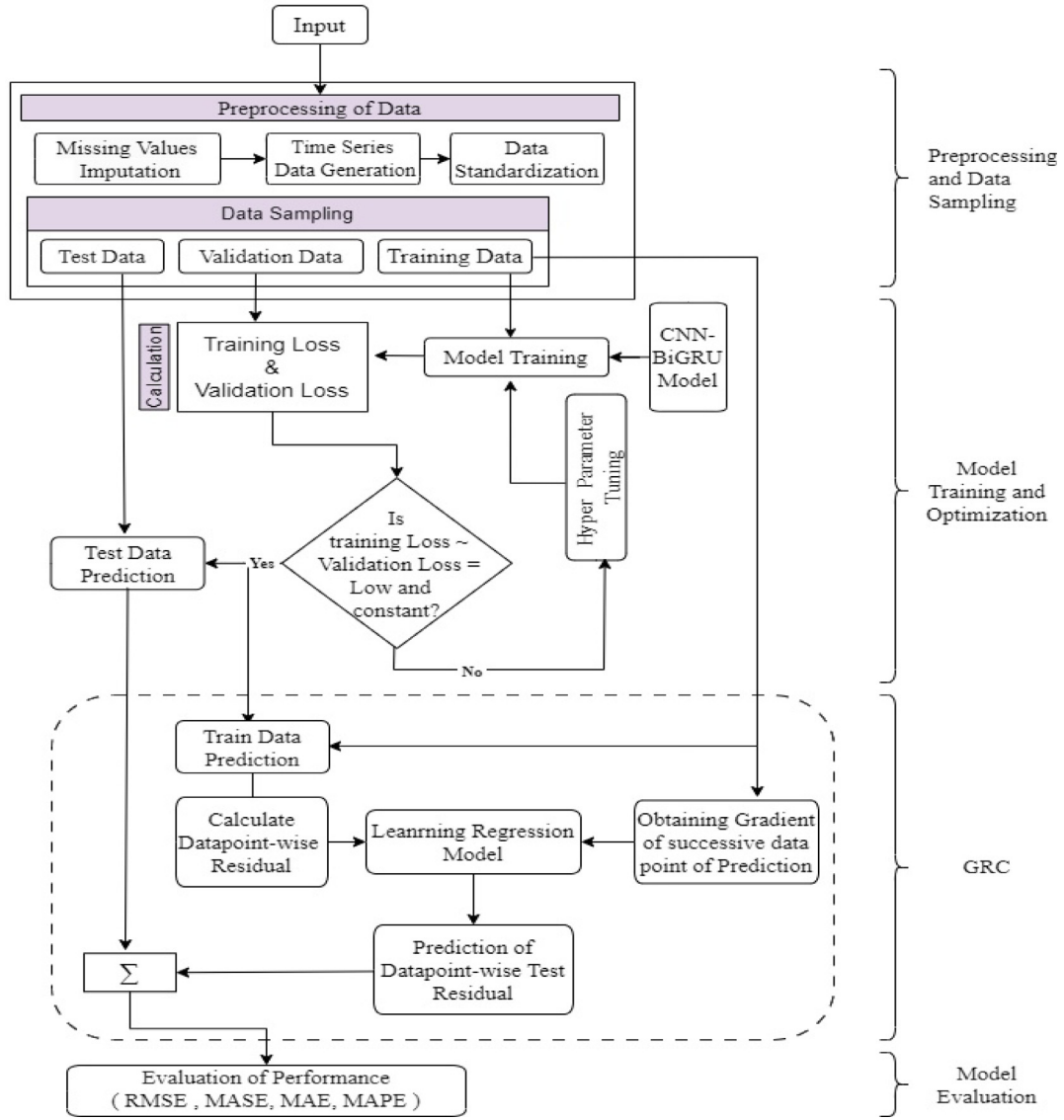


Fig. 4. GRC technique(Proposed 2) along with CNN-BiGRU (Proposed 1) for improved prediction of SSN.

The Bi-GRU is a two-way GRU that has two directions, namely Forward GRU and Backward GRU, and corresponding equations related to them.

Forward GRU:

$$r_t^f = \sigma(W_{xr}^f x_t + W_{hr}^f h_{t-1}^f + b_r^f) \quad (5)$$

$$z_t^f = \sigma(W_{xz}^f x_t + W_{hz}^f h_{t-1}^f + b_z^f) \quad (6)$$

$$\tilde{h}_t^f = \tanh(W_{xh}^f x_t + r_t^f \odot (W_{hh}^f h_{t-1}^f) + b_h^f) \quad (7)$$

$$h_t^f = (1 - z_t^f) \odot h_{t-1}^f + z_t^f \odot \tilde{h}_t^f \quad (8)$$

Eq. 5 is the reset gate for the forward GRU with σ as the sigmoid activation function. W_{hr}^f is the weight matrices connecting the hidden state h_{t-1}^f of forward GRU unit to its reset gate r_t^f . It is the transformation from a hidden state of the previous time step to the reset gate in the forward direction. Similarly, W_{xr}^f is the weight matrix associ-

ated with the input in the current time step x_t to the reset gate. b_r^f is the bias term associated with this gate. Eq. 6 represents the equation for the update gate in forward GRU with W_{xz}^f and W_{hz}^f as weight matrices associated with it, which represents weight connecting the input of current time step to the update gate and weight connecting hidden state of previous time step to update gate respectively. b_z^f is the bias term associated with the update gate. In the Eqs. 7 and 8, which represent the equation for hidden state for time step t and candidate activation, respectively, the symbol $\odot$ represents the element-wise multiplication and $\tanh$ is the hyperbolic tangent activation function. The weights W_{xh}^f and W_{hh}^f signify weight matrices connecting the input and hidden state of the previous time step to the candidate activation.

Backward GRU:

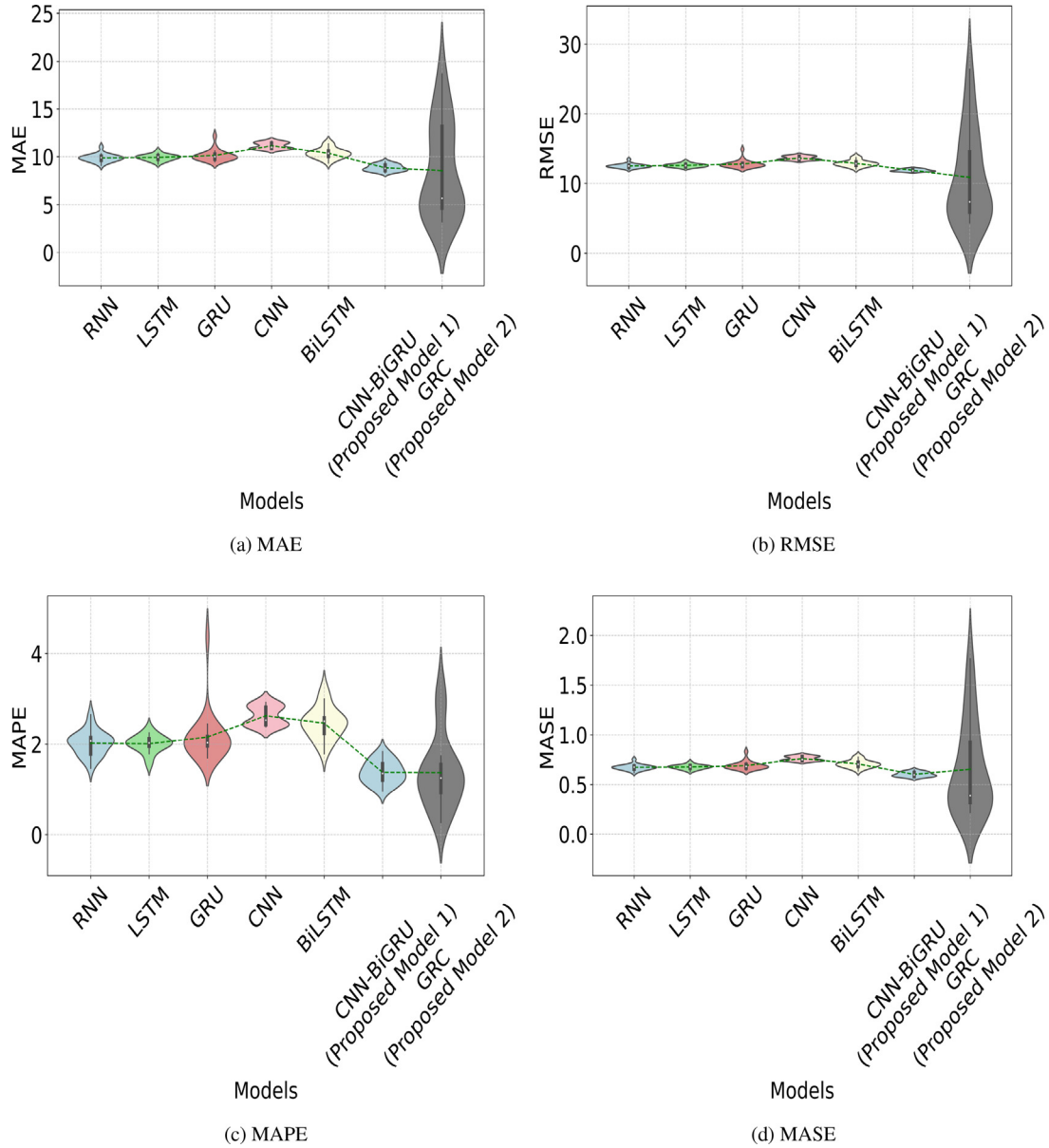


Fig. 5. Performance Measures of traditional models along with Proposed Model 1 and Proposed Model 2 over the daily SSN.

$$r_t^b = \sigma(W_{xr}^b x_t + W_{hr}^b h_{t+1}^b + b_r^b) \quad (9)$$

$$z_t^b = \sigma(W_{xz}^b x_t + W_{hz}^b h_{t+1}^b + b_z^b) \quad (10)$$

$$\tilde{h}_t^b = \tanh(W_{xh}^b x_t + r_t^b \odot (W_{hh}^b h_{t+1}^b) + b_h^b) \quad (11)$$

$$h_t^b = (1 - z_t^b) \odot h_{t+1}^b + z_t^b \odot \tilde{h}_t^b \quad (12)$$

The backward GRU is similar to the forward GRU but in the reverse direction. Eq. 9 represents the reset gate with W_{xr}^b and W_{hr}^b as the weight matrices connecting input to reset gate and hidden state from one time step ahead to the reset gate in the backward direction. Similarly, Eq. 10 represents the equation for the update gate, where W_{xz}^b and W_{hz}^b signify the weight matrix for the input in the current time step to the update gate and the hidden state of a one-time step ahead to the update gate of backward GRU,

respectively. Eqs. 11 and 12 represent the equations for the hidden state of backward GRU and candidate activation at time step t respectively where W_{xh}^b and W_{hh}^b in Eq. 11 represents weight matrices relating input to candidate activation and connecting hidden state of one-time step ahead to candidate activation respectively.

Concatenation:

$$h_t = [h_t^f, h_t^b] \quad (13)$$

In Eq. 13, h_t represents the hidden layer at time step t , which is a combination of both hidden states in forward and backward GRU. Apart from these, the study also includes traditional DL models, namely LSTM, GRU, BiLSTM, CNN, and RNN, for comparison with the proposed model.

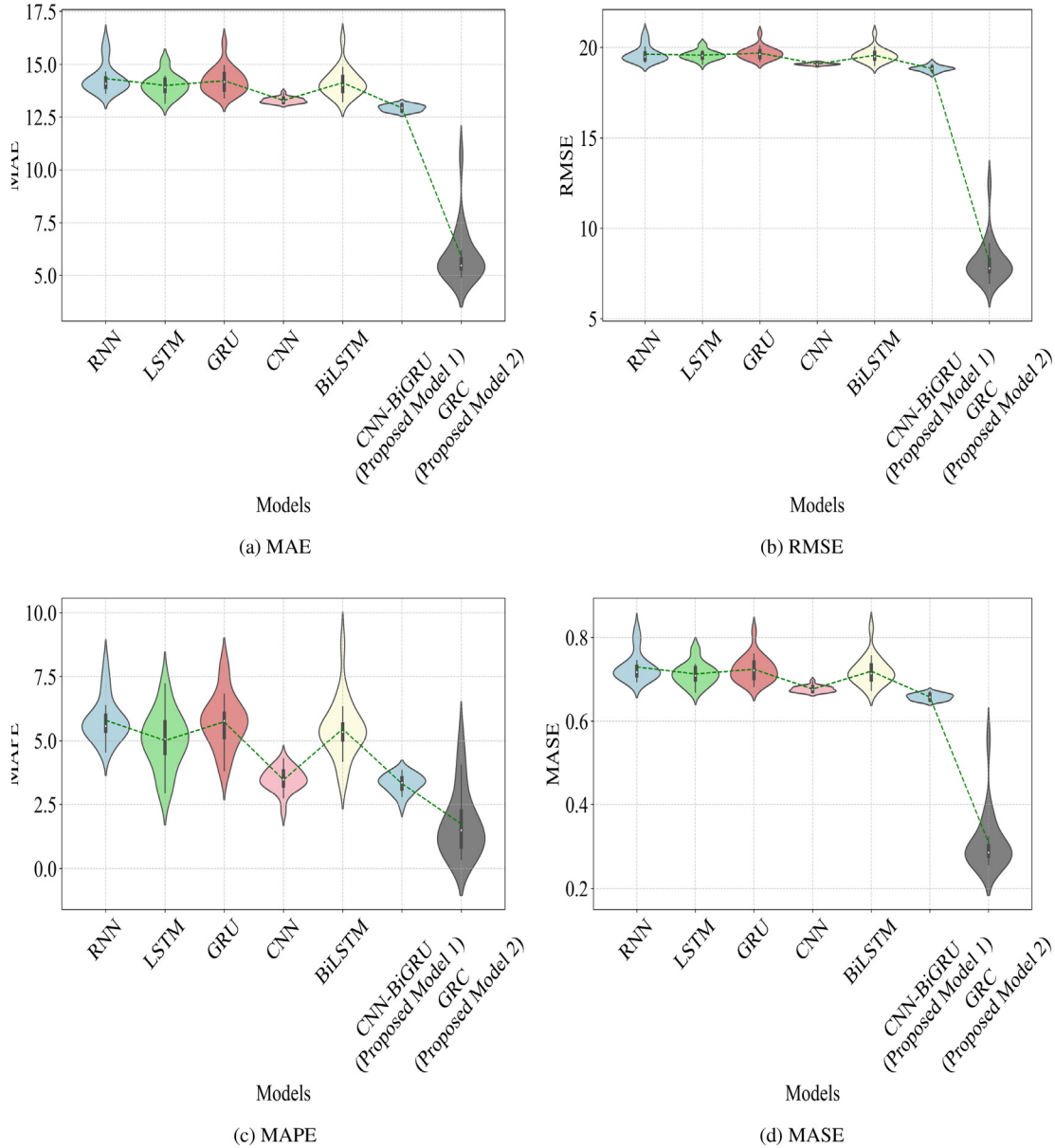


Fig. 6. Performance Measures of traditional models along with CNN-BiGRU (Proposed 1) and GRC (Proposed 2) models over the monthly mean SSN.

3.4. Proposed model 2: gradient residual correction

This is a proposed post-processing technique where the prediction results obtained from the proposed model have been improved using the prediction of the residual component of the test data based on the residual component of the training set and related gradient of the training data using AdaBoost regression model with 100 estimators. The result thus obtained is summed up with earlier obtained predictions, and performance analysis validated the significant improvement in bringing down the error in the predictions. The working of the AdaBoost regressor is as follows:

In Eq. 14, $w_i^{(1)}$ is the weight of the i^{th} training sample in the first iteration, and n represents the number of samples. After fitting a weak regressor $h_t(x)$ to the training data using the weights $w_i^{(t)}$, we have Eq. 15 which represents a weighted error of weak regressor ϵ_t . The $h_t(x)$ represents the output of the weak regressor at iteration t for input x , and y_i represents the target value of the i^{th} training sample.

$$\epsilon_t = \sum_{i=1}^n w_i^{(t)} |h_t(x_i) - y_i| \quad (15)$$

$$\alpha_t = \frac{1}{2} \ln \left(\frac{1 - \epsilon_t}{\epsilon_t} \right) \quad (16)$$

$$w_i^{(1)} = \frac{1}{n}, \quad i = 1, 2, \dots, n \quad (14)$$

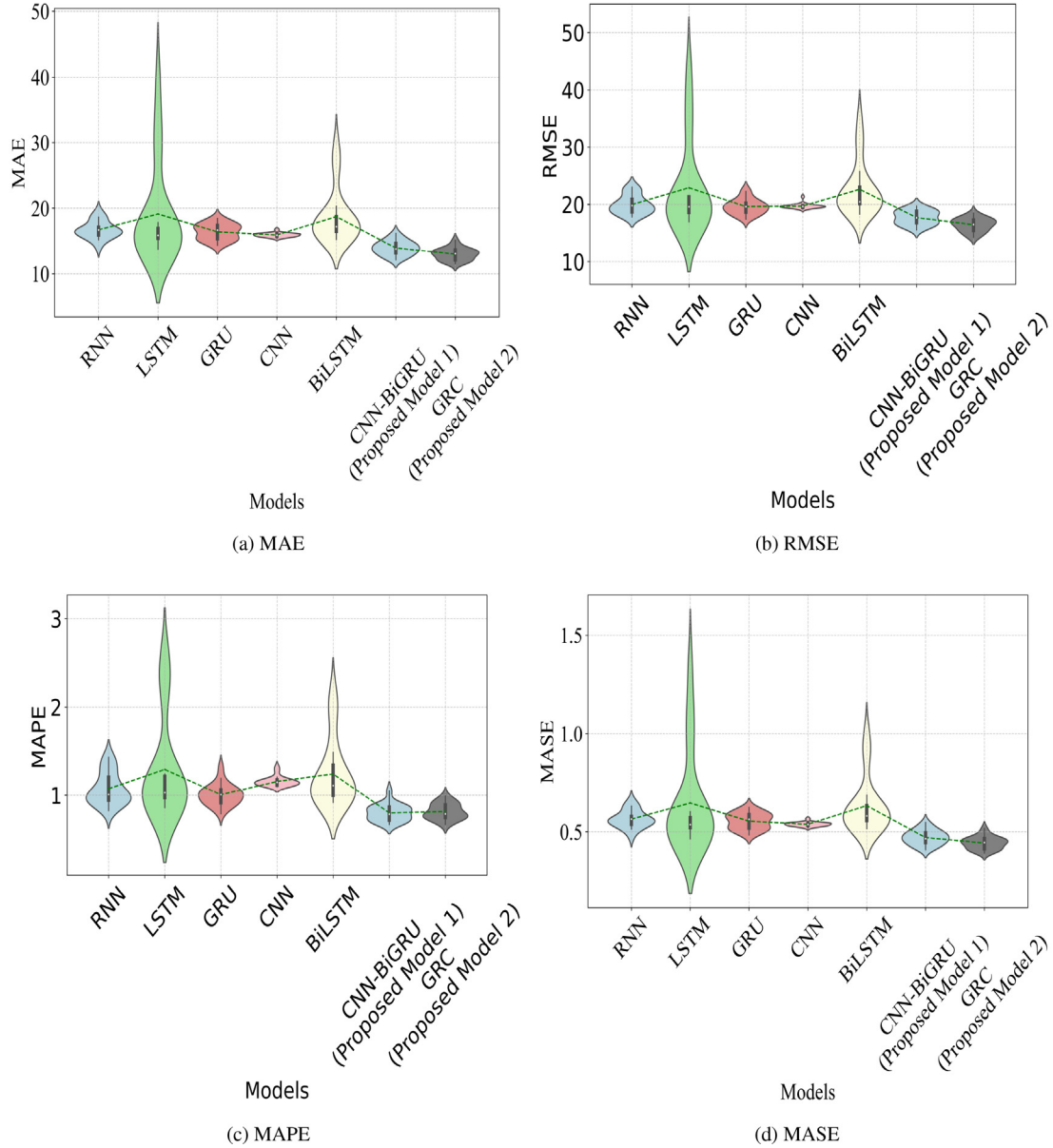


Fig. 7. Performance Measures of traditional a models along with CNN-BiGRU (Proposed 1) and GRC (Proposed 2) models over the yearly mean SSN.

Eq. 16 represents the computed weight of the weak regressor as α_t . The weight update of the training samples takes place as depicted in Eq. 17:

$$w_i^{(t+1)} = \frac{w_i^{(t)} \exp(-\alpha_t \cdot y_i \cdot h_t(x_i))}{Z_t} \quad (17)$$

Then normalization factor Z_t is calculated as depicted in the Eq. 18

$$Z_t = \sum_{i=1}^n w_i^{(t)} \exp(-\alpha_t \cdot y_i \cdot h_t(x_i)) \quad (18)$$

Finally, the ensemble prediction $\hat{y}(x)$ is calculated as presented in Eq. 19, where T is the number of weak estimators.

$$\hat{y}(x) = \sum_{t=1}^T \alpha_t \cdot h_t(x) \quad (19)$$

In the post-processing phase, the CNN-BiGRU (Proposed 1) is trained and utilized for predicting the values for training data itself which is $y_{tr} \in D'_{tr}$ producing a set of predicted values $\hat{y}_{tr}$ for training data. Using Eq. 20, training residual $\hat{y}_i^{res}$ is calculated as:

$$[\hat{y}_i^{res}]_{i=1}^n = [y_{tr} - \hat{y}_{tr}]_{i=1}^n \quad \forall y_{tr} \in D'_{tr}, \hat{y}_{tr} \in D''_{tr}, \hat{y}_i^{res} \in D_y^{res} \quad (20)$$

where D''_{tr} is the training prediction set and D_y^{res} is the set of residuals obtained from training dataset D'_{tr} and its corresponding predictions D''_{tr} . The gradient of training data

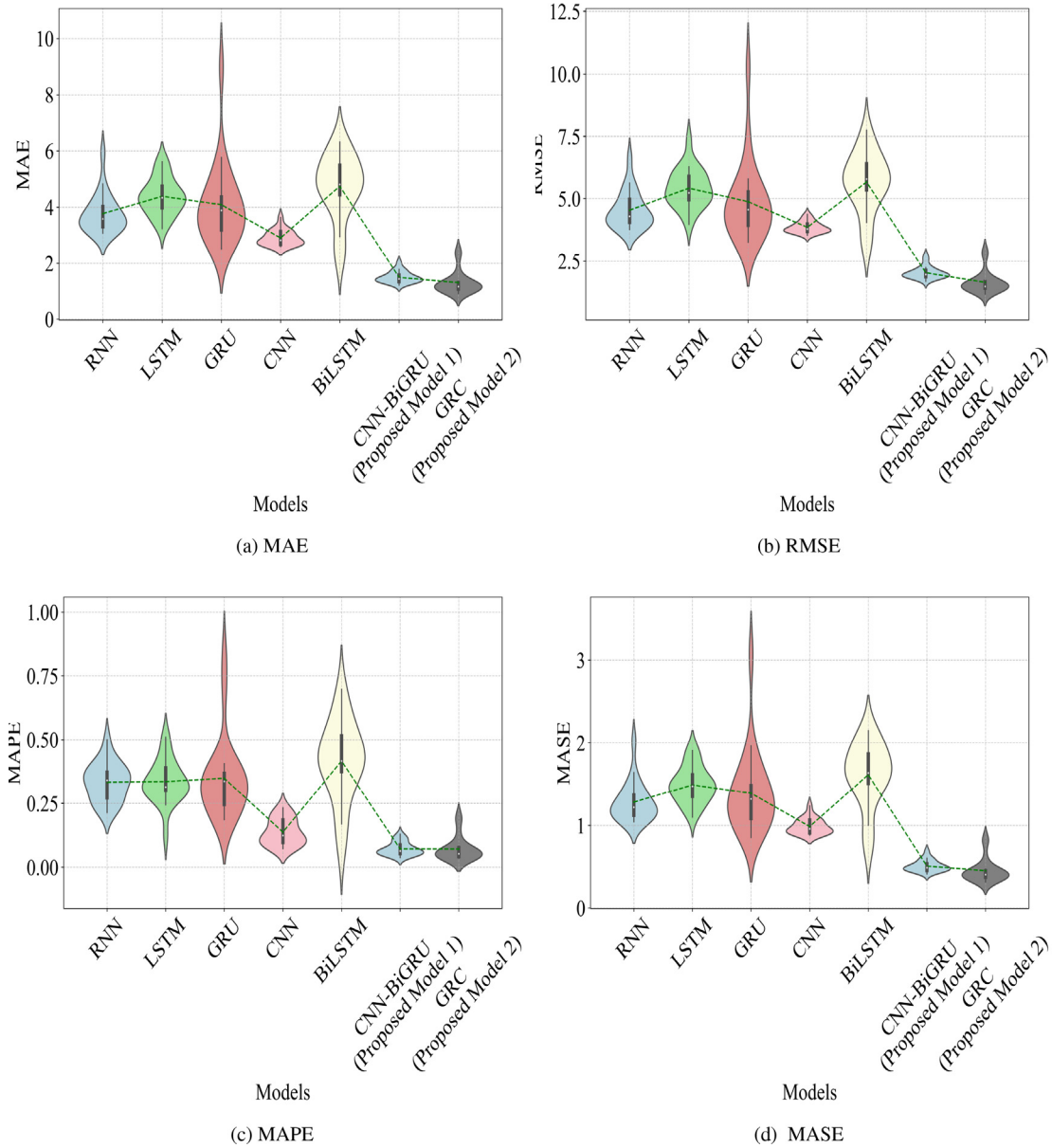


Fig. 8. Performance Measures of traditional models along with CNN-BiGRU (Proposed 1) and GRC (Proposed 2) models over the 13-month smoothed SSN.

y_i^{grad} at every datapoint is calculated with respect to its previous time step as represented in Eq. 21

$$[y_i^{grad}]_{i=1}^n = [y_i - y_{i-1}]_{i=1}^n \quad \forall y_i \in D'_{tr}, y_i^{grad} \in D_y^{grad} \quad (21)$$

where D_y^{grad} is the dataset of gradient values. Then the values of D_y^{res} and D_y^{grad} are fed to the AdaBoost Regressor as X and Y for training the regression model. The resultant trained model is then utilized to predict residual for test data. The gradient of the test predictions D_y^{grad} obtained from the predicted test dataset $\hat{y}_{te}$, which has already been obtained from the proposed hybrid DL model as test data predictions are fed to the trained AdaBoost Regressor model to obtain residual for the correction of test prediction. The result obtained from the AdaBoost Regressor

model is termed as the residual component y_{te}^{res} , which is added with the previously predicted test values obtained from the hybrid DL model, producing the final predictions $\hat{y}_i^F$ as depicted in the Eq. 22

$$[\hat{y}_i^F]_{i=1}^n = [\hat{y}_{te} + \hat{y}_i^{res}]_{i=1}^n \quad \forall \hat{y}_i^F \in D_y'', \hat{y}_{te} \in D_{Te}'', \hat{y}_i^{res} \in D_y^{res} \quad (22)$$

where D_y'' is the set of final predictions, D_{Te}'' is the test data predictions obtained from CNN-BiGRU (Proposed 1) model and D_y^{res} is the set of residual for the predicted test dataset. The four datasets are used for training the model individually. All four datasets are univariate, so these were converted into time-series data with a lag of 11, resulting in the data being converted into a two-dimensional matrix.

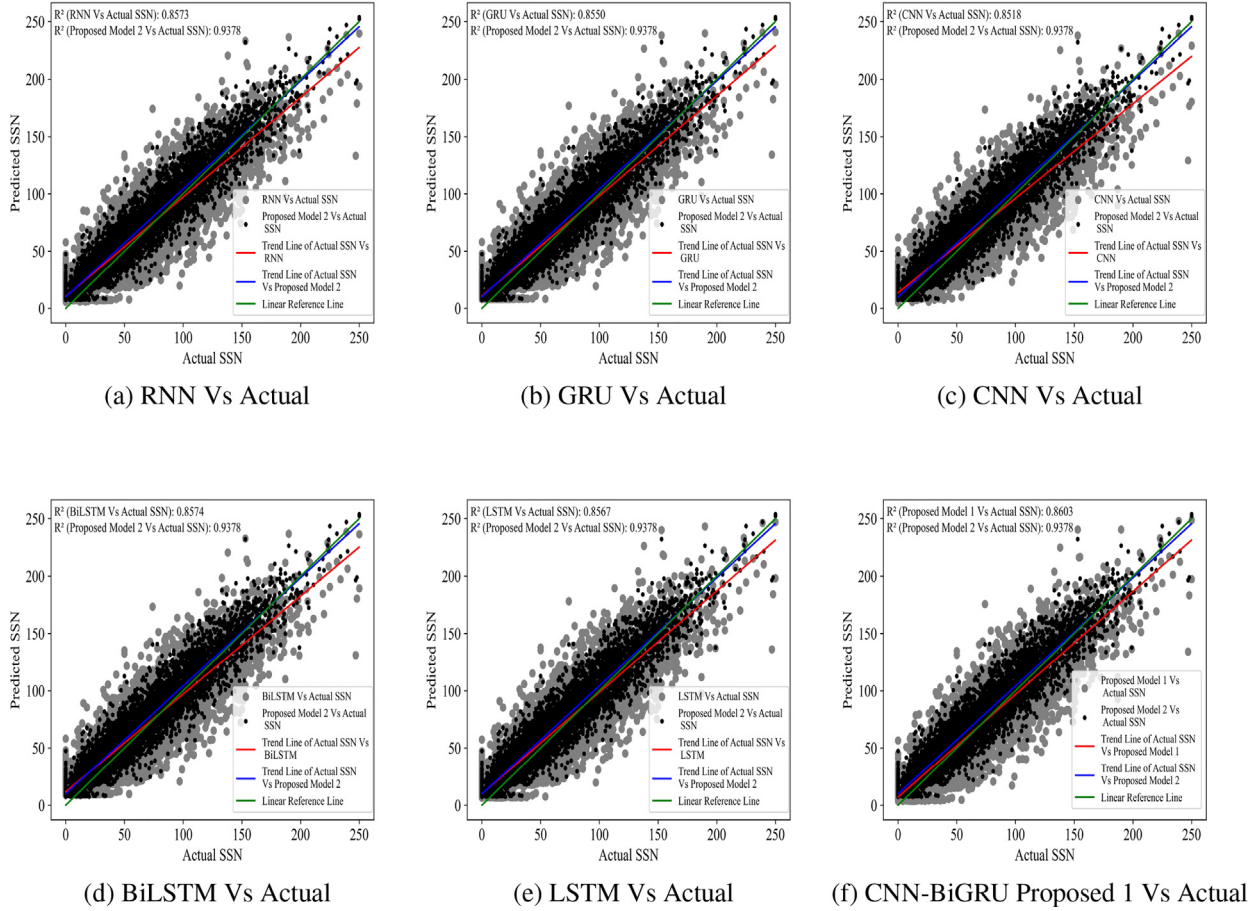


Fig. 9. Comparison of traditional and CNN-BiGRU (Proposed 1) model vs actual superposed over GRC (Proposed 2) model vs actual on the daily SSN.

For both monthly variants, it became a group of one-year data with internal seasonality, while for the annual variant, it took the shape of one complete solar cycle as one solar cycle is comprised of 11 years. At every learning step, one dimension of this matrix is passed to the layers of the model, where 11 values train the model for the one resultant value. After training the model, it is used to predict the training samples. The residual for each prediction, which the model made for the training data, is recorded and later utilized as the training data for the AdaBoost regressor. When the Proposed Model 1 predicted the test data, the AdaBoost regressor model, which was trained to predict the residual, was utilized to predict the residuals for test data. Since the test data predicted by the Proposed Model 1 and corresponding predicted residuals predicted by the AdaBoost regressor are present at this stage, they are aggregated to determine the final predictions.

3.4.1. Performance metrics

The study utilizes four performance metrics, namely RMSE, MASE, MAPE, and MAE, which include both data-dependent and independent measures for better comparison. Most of the studies analyzed in the previously existing literature also follow similar evaluation parameters as depicted in Table 1.

These measures depict the deviation of the predicted result with the actual values from different aspects. Prior to utilizing the aforementioned evaluation parameters, the concept of **residual error** is essential, which is the difference between y and $\hat{y}$ where y is the original value of SSN and $\hat{y}$ is the forecasted value of SSN in the context of these experiments. The equations related to the RMSE are represented in Eq. 23 where k is the number of samples along with y_j , and $\hat{y}_j$ are the j^{th} observed and predicted value respectively.

$$RMSE = \sqrt{\frac{1}{k} \sum_{j=1}^k (y_j - \hat{y}_j)^2} \quad (23)$$

Eq. 24 illustrates the formula of Mean Absolute error, which is the common measure for performance evaluation. In this equation, D represents the complete test dataset.

$$MAE = \sum_{j=1}^D |(y_j - \hat{y}_j)| \quad (24)$$

Eq. 25 illustrates the equation of Mean absolute scaled error where e_k represents forecast error, also known as residual error, s is the seasonal period, and T corresponds to the total count of data points contained within the time

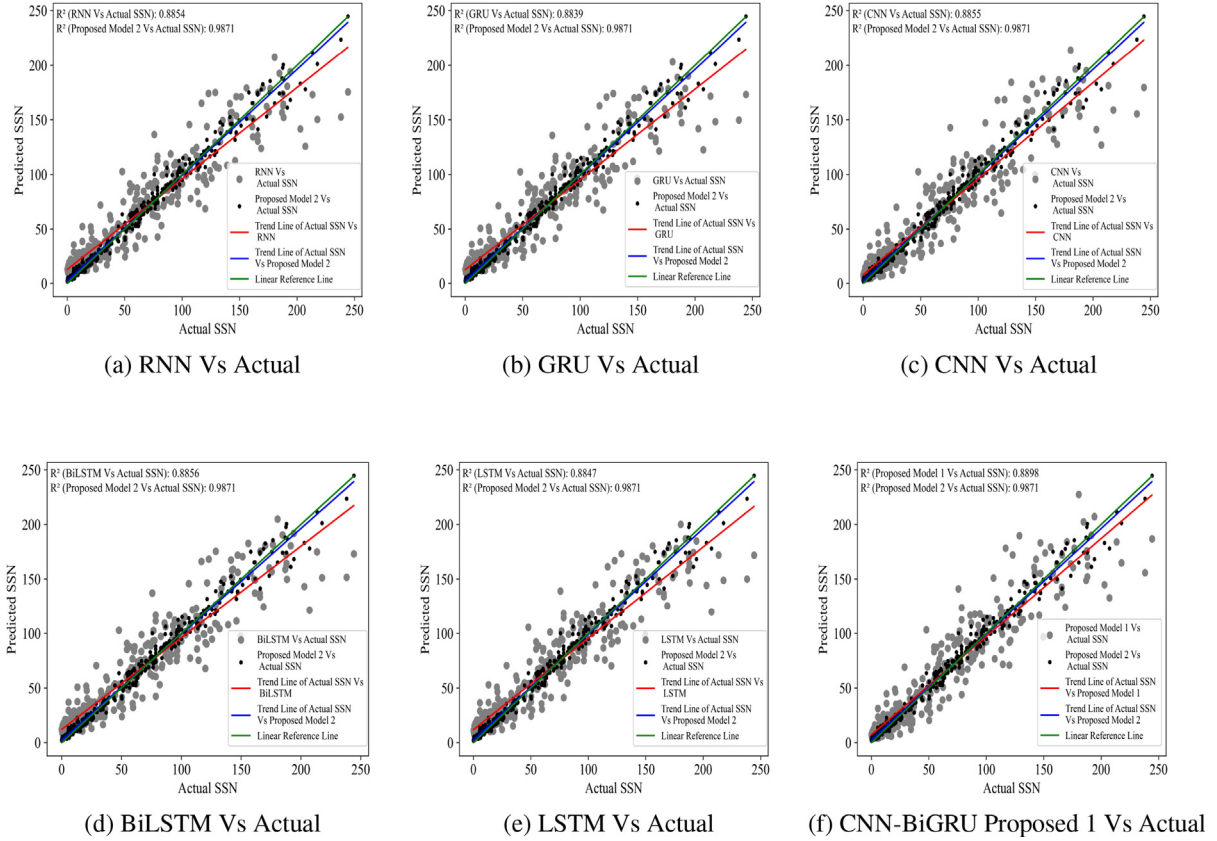


Fig. 10. Comparison of traditional and CNN-BiGRU (Proposed 1) model vs actual superposed over GRC (Proposed 2) model vs actual over the monthly mean SSN.

series data, and the summation part in the denominator is the mean absolute error.

$$MASE = \text{mean} \left(\frac{|e_k|}{\frac{1}{T-s} \sum_{t=s+1}^T |Y_t - Y_{t-s}|} \right) \quad (25)$$

MAPE is a data-independent model evaluation parameter, which is used to calculate the error in the context of percentage. The associated formula is presented in Eq. 26, where A_j refers to the actual and F_j refers to the predicted value.

$$MAPE = \frac{1}{k} \sum_{j=1}^k \left| \frac{A_j - F_j}{A_j} \right| \quad (26)$$

4. Results analysis and discussion

In this section, the observations obtained from the execution of traditional DL models, namely LSTM, GRU, CNN, RNN, and BiLSTM, are compared with the hybrid DL model CNN-BiGRU (Proposed Model 1) and GRC (Proposed Model 2) technique. LSTM, GRU, and RNN models used in this study consist of 16 units each. BiLSTM

model consists of 8 units of LSTM, whereas CNN consists of the one-dimensional convolutional layer with a filter size of 16 and kernel size equal to 1. All models utilized 'relu' as the activation function. LSTM, GRU, BiLSTM, and RNN contain their single recurrent layer followed by dropout and a fully connected layer with a single unit. Similarly, CNN consists of a single one-dimensional convolutional layer followed by the dropout and fully connected layer with a single unit. All models are trained using the 'Adam' optimizer, and batch size is equal to 66. The number of epochs for each model is equal to 250, with early stopping criteria enabled while monitoring validation loss with a patience of 20. The hyperparameter associated with the CNN-BiGRU (Proposed Model 1) model has already been discussed in Section 3.3.1.

The MAE obtained for different models and improvement obtained in the MAE after the implementation of the GRC (Proposed Model 2) is depicted in Fig. 5a. One violin represents 20 independent MAE obtained from 20 different runs of the individual model over one dataset. It can be observed from Fig. 5 that the proposed framework has performed better than all other traditional DL approaches on the daily SSN data overall performance measures except for MASE. The variance in the results of the Proposed Model 2 over all other models is high for the daily SSN data because the daily SSN data has a higher

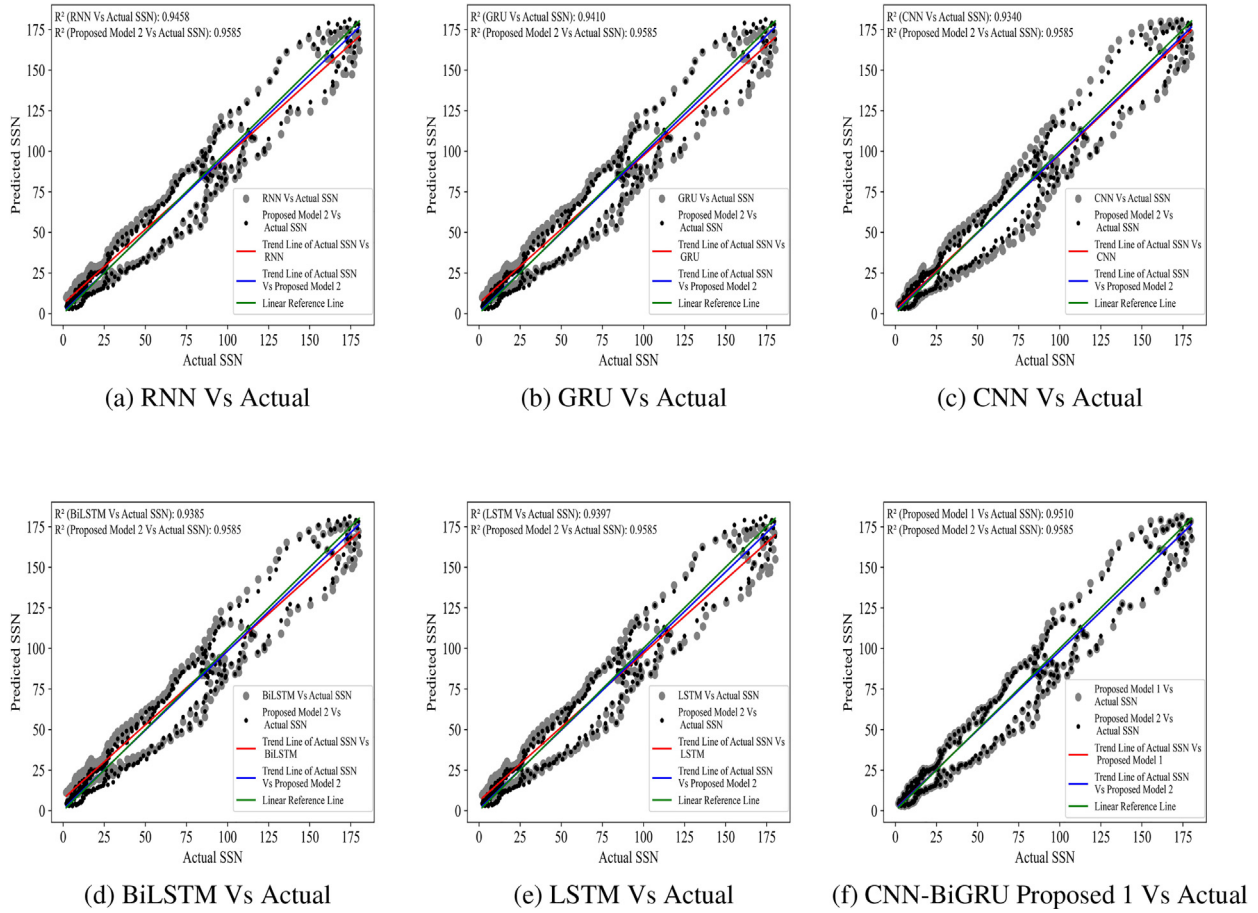


Fig. 11. Comparison of traditional and CNN-BiGRU (Proposed 1) model vs actual superposed over GRC (Proposed 2) model vs actual over the monthly smooth SSN.

number of test samples with more fluctuations, and the Proposed Model 2 is based on regression learning. Many test samples with high fluctuations reduce the correlation between the gradient and the corresponding residual. Therefore, the variance in the prediction may yield an increment in the error component. Fig. 5 also depicts that CNN-BiGRU (Proposed Model 1) has also performed better than all other traditional models on daily SSN over MAE, RMSE, and MAPE. The effectiveness of the proposed model is enhanced by the GRC (Proposed Model 2). Fig. 6 depicts the performance of all models in comparison to the proposed methods over the monthly average SSN data. A significant improvement in the effectiveness can be observed on all experimented evaluation parameters. The CNN-BiGRU (Proposed Model 1) model is performing better than the traditional DL approaches. The implementation of the GRC (Proposed Model 2) improved the effectiveness to a significant level. The fluctuations in the test data are low since the monthly mean SSN data has fewer test samples. Due to this, the AdaBoost regressor present in the Proposed Model 2 was trained effectively. This training yielded a better residual prediction for the test data and improved overall accuracy with less variance. Fig. 7 represents the performance of all experimented mod-

els over the yearly average SSN data, and it is found that both the proposed methods CNN-BiGRU (Proposed Model 1) individually and in association with the GRC (Proposed Model 2) have improved the performance on all evaluation parameters included in the experiment. Fig. 8 illustrates the comparative analysis of CNN-BiGRU (Proposed Model 1) and GRC (Proposed Model 2) with all other traditional approaches. The improved effectiveness can also be observed in this case on all performance measures. The number of test samples in this dataset is similar to the number of test samples in the monthly mean SSN data. In addition, this data is already smoothed, as described in Section 2, so the fluctuations are much lower than the other three variants of the data. Due to fewer fluctuations, the AdaBoost regressor in Proposed Model 2 predicted the residual more effectively. The effective prediction of residual resulted in better final predictions with less variance.

Thus, it can be deduced that the proposed methods are good for both monthly and annual frequency of SSN data. It also has an effective performance over the daily SSN data. Fig. 9 depicts the scatter plot of the actual vs predicted daily SSN of the proposed framework superimposed over the actual daily SSN vs predicted daily SSN of the dif-

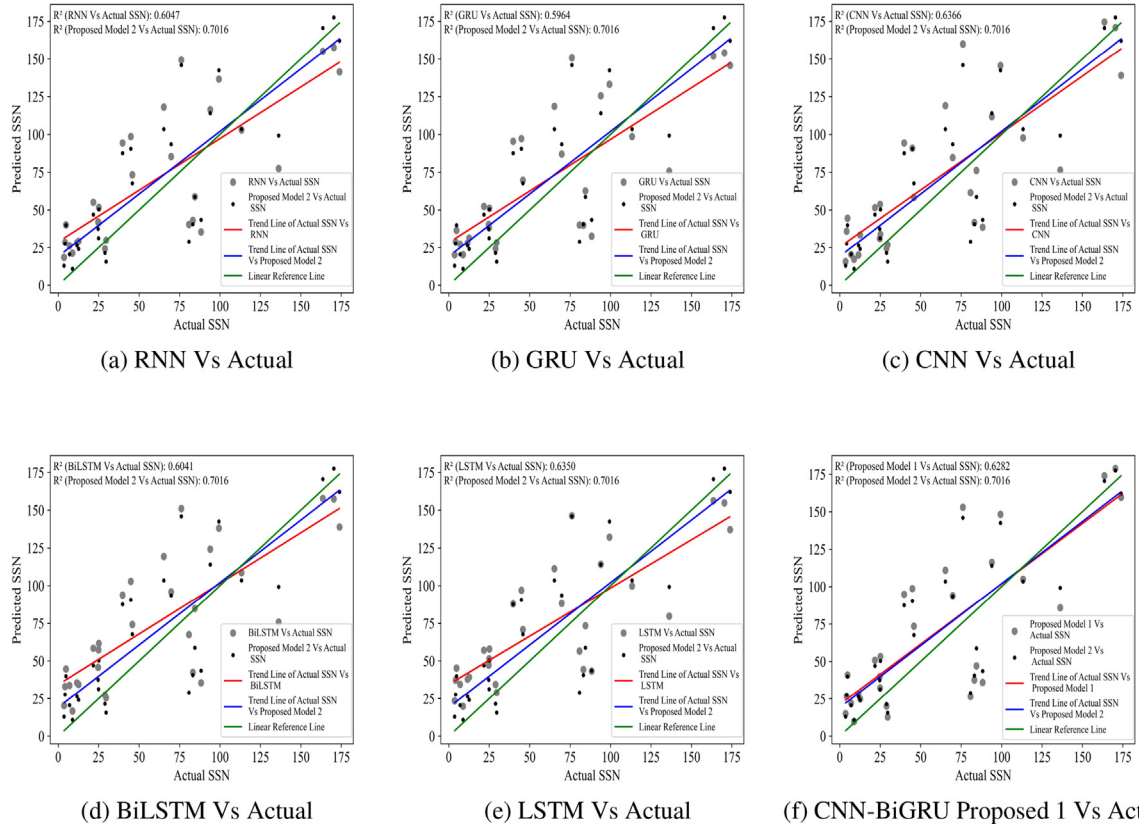


Fig. 12. Comparison of traditional and CNN-BiGRU model (Proposed Model 1) vs actual superposed over GRC (Proposed Model 2) vs actual over the yearly mean SSN.

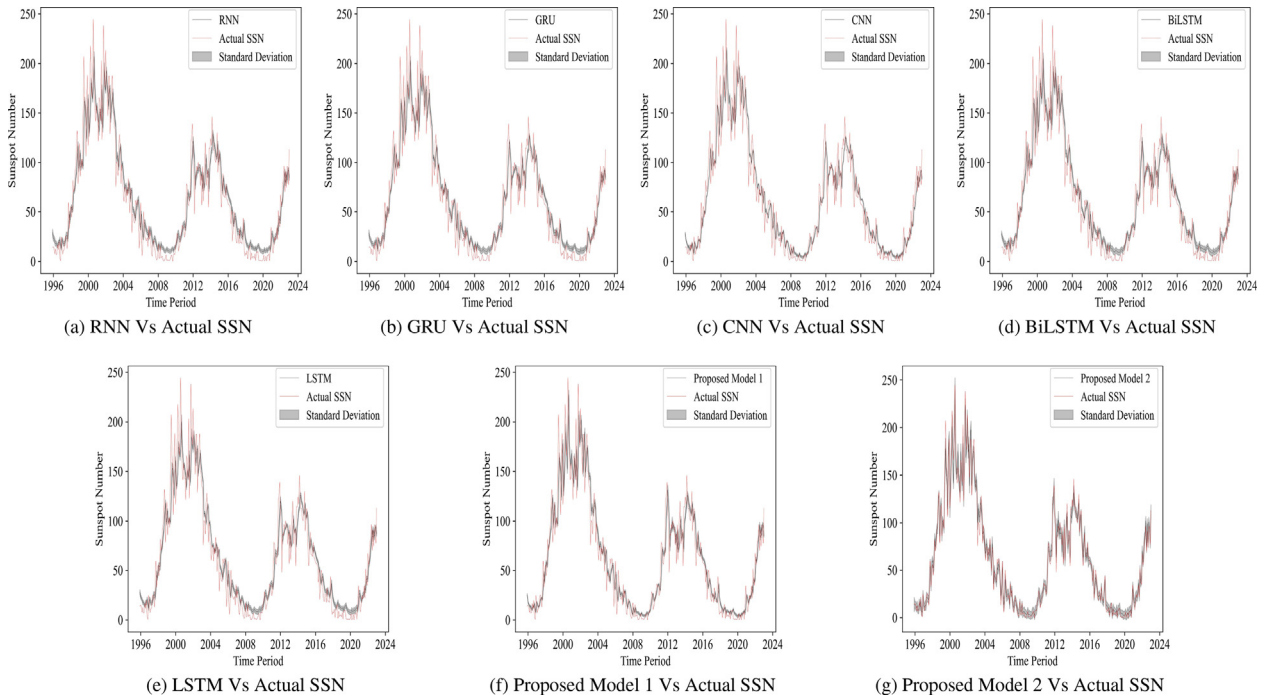


Fig. 13. Comparison of prediction of all models with respect to actual over the monthly mean SSN.

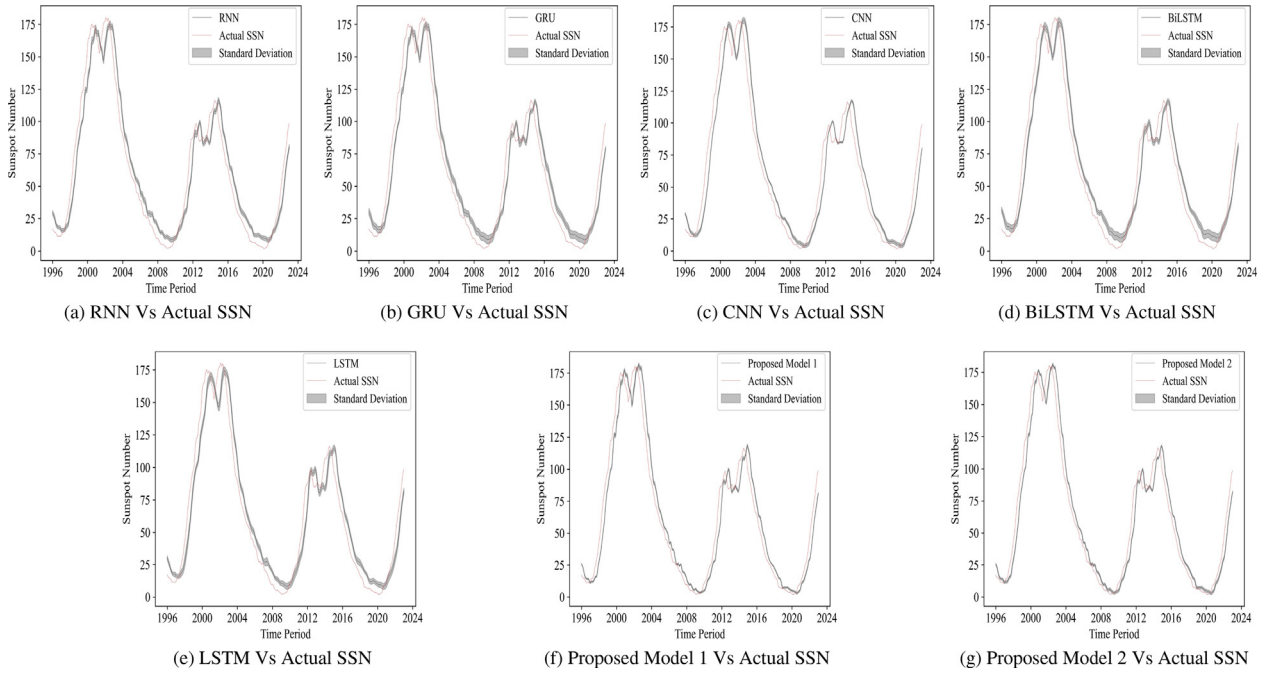


Fig. 14. Comparison of prediction of all models with respect to actual over the monthly smooth SSN.

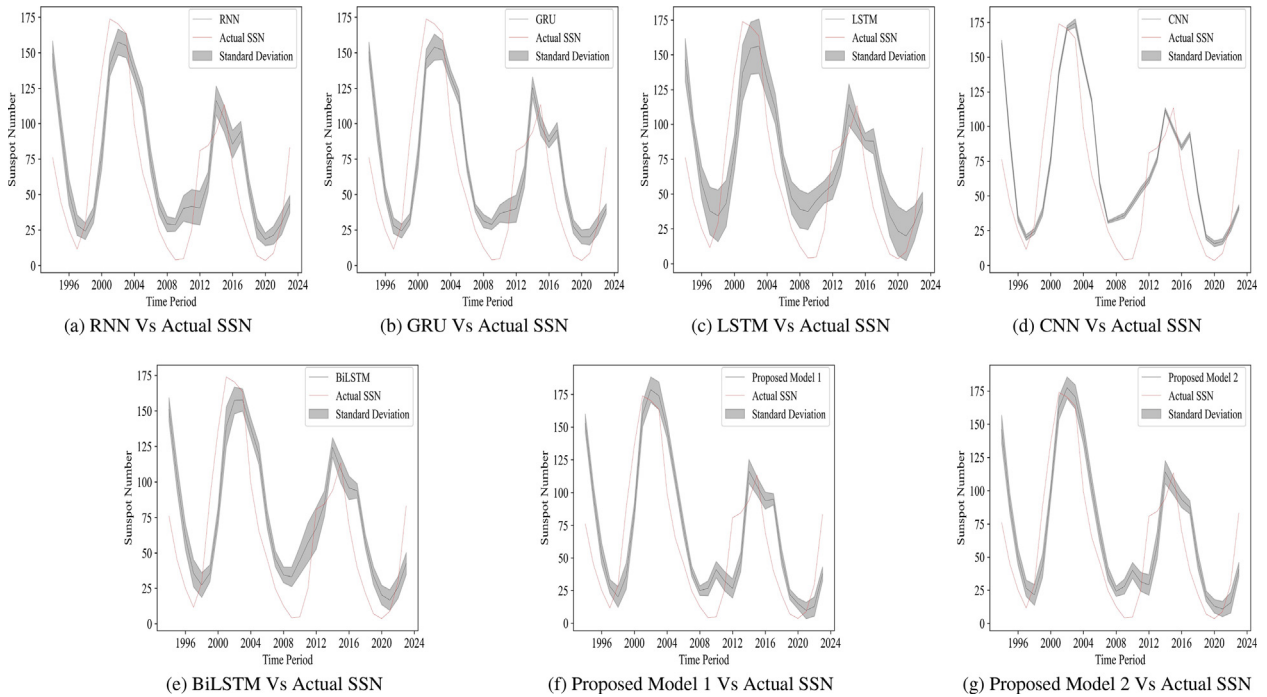


Fig. 15. Comparison of prediction of all models with respect to actual over the yearly mean SSN.

ferent traditional model and hybrid CNN-BiGRU model (Proposed Model 1). These results are for the average of 20 predictions from 20 different runs of each model. The closeness of the superposed plot to the linear reference line of 45 degrees to the scatter plot of other models illustrates that the GRC (Proposed Model 2) technique has a better prediction over each experimented model. The R^2 value

of the GRC (Proposed Model 2) depicts the same. The R^2 value for the proposed framework is 0.9378. It is higher than the R^2 score of other models, as depicted in Fig. 9c. The trend line for models is also included for comparative analysis. Fig. 10 illustrates similar results over the monthly average SSN. In this observation, the R^2 value of the proposed framework is 0.9871, which is better than all other

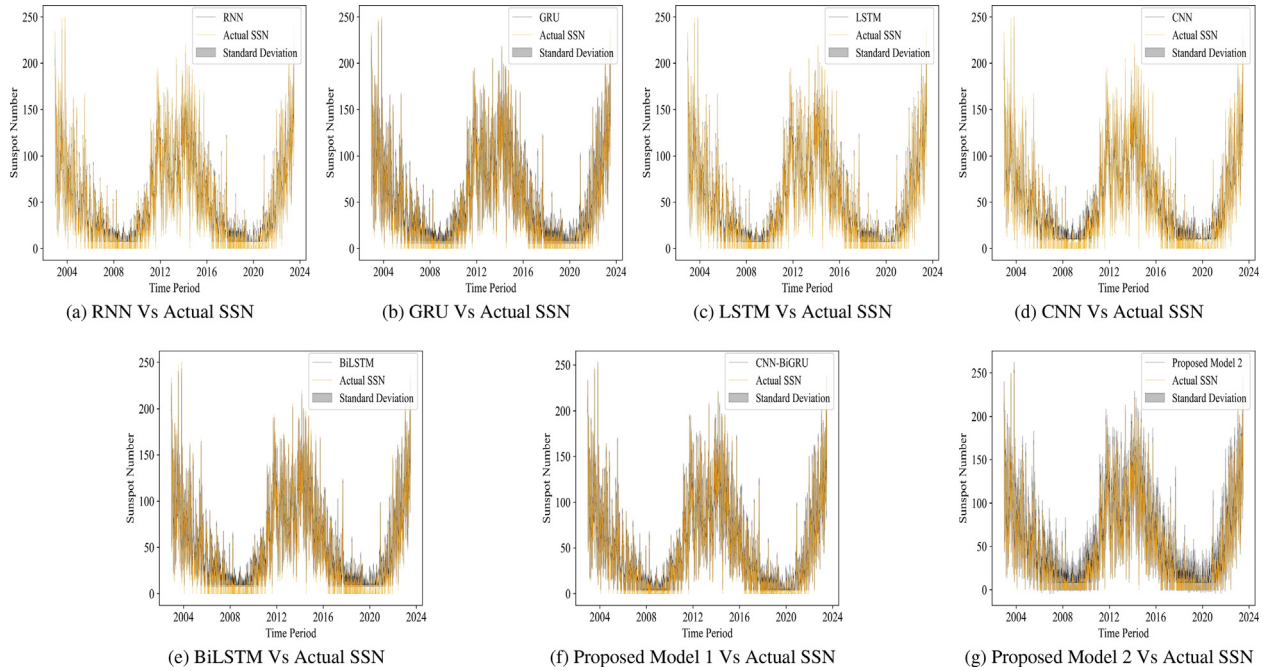


Fig. 16. Comparison of prediction of all models with respect to actual over the daily SSN.

Table 3

Performances of traditional and proposed models, i.e. CNN-BiGRU and GRC over the four datasets.

Dataset	Models	RMSE	MAE	MASE	MAPE
Yearly Mean SSN	LSTM	22.87 (+28.03%)	19.11 (+32.03%)	0.65 (+33.3%)	1.29 (+37.21%)
	GRU	19.62 (+16.11%)	16.39 (+20.74%)	0.55 (+20.0%)	1.00 (+19.00%)
	CNN	19.70 (+16.45%)	15.93 (+18.45%)	0.54 (+18.51%)	1.16 (+30.17%)
	RNN	19.98(+17.62%)	16.70 (+22.22%)	0.56 (+21.43%)	1.07 (+24.30%)
	BiLSTM	22.61(+27.20%)	18.72 (+30.61%)	0.63 (+30.16%)	1.24 (+34.68%)
	CNN-BiGRU (Proposed 1)	17.70 (+7.00%)	13.94 (+6.81%)	0.47 (+6.38%)	0.80 (-1.25%)
	GRC (Proposed Model 2)	16.46	12.99	0.44	0.81
Monthly Mean SSN	LSTM	19.57(+58.30%)	14.00 (+57.71%)	0.71 (+56.34%)	5.01 (+65.47%)
	GRU	19.68 (+58.54%)	14.22 (+58.37%)	0.72(+56.94%)	5.74(+69.86%)
	CNN	19.08 (+57.23%)	13.30(+55.49%)	0.68 (+54.41%)	3.47(+50.14%)
	RNN	19.63 (+58.43%)	13.93(+57.50%)	0.73(+57.53%)	5.79 (+70.12%)
	BiLSTM	19.57 (+58.30%)	14.14 (+58.13%)	0.72 (+56.94%)	5.46 (+68.32%)
	CNN-BiGRU (Proposed 1)	18.84 (+56.69%)	12.92 (+54.18%)	0.66 (+53.03%)	3.32(+47.89%)
	GRC (Proposed Model 2)	8.16	5.92	0.31	1.73
13- month smooth SSN	LSTM	5.42 (+69.74%)	4.38 (+70.09%)	1.49 (+69.80%)	0.33 (+81.81%)
	GRU	4.88 (+66.39%)	4.08 (+67.89%)	1.39 (+67.63%)	0.35 (+82.86%)
	CNN	3.85 (+57.40%)	2.90 (+54.83%)	0.99 (+54.54%)	0.14 (+57.14%)
	RNN	4.54 (+63.88%)	3.77 (+65.25%)	1.28 (+64.84%)	0.33 (+81.81%)
	BiLSTM	5.66 (+71.02%)	4.74 (+72.36%)	1.61 (+72.05%)	0.42 (+ 85.71%)
	CNN-BiGRU (Proposed 1)	2.03 (+19.21%)	1.50 (+12.67%)	0.51 (+11.76%)	0.07 (+14.28%)
	GRC (Proposed Model 2)	1.64	1.31	0.45	0.06
Daily SSN	LSTM	12.58 (+13.67%)	9.91 (+13.82%)	0.67 (+2.98%)	2.00 (+32.00%)
	GRU	12.77 (+14.96%)	10.12 (+15.61%)	0.69 (+5.80%)	2.16(+37.04%)
	CNN	13.64 (+20.38%)	11.16 (+23.48%)	0.76(+14.47%)	2.63(+48.29%)
	RNN	12.5 (+13.12%)	9.86 (+13.39%)	0.67 (+2.99%)	2.02(+32.67%)
	BiLSTM	12.87 (+15.62%)	10.36 (+17.57%)	0.71(+8.45%)	2.46(+44.72%)
	CNN-BiGRU (Proposed 1)	11.88 (+8.59%)	8.83(+3.28%)	0.6 (-8.33%)	1.38 (+1.45%)
	GRC (Proposed Model 2)	10.86	8.54	0.65	1.36

experimented traditional models, including CNN, with a R^2 value equal to 0.8855. It also improved the performance of the proposed hybrid CNN-BiGRU model (Proposed Model 1) from a R^2 value of 0.8898 to 0.9871. The scatter

plot of actual vs predicted SSN for the 13-month smoothed dataset is represented in Fig. 11. In this case, the R^2 value obtained from the GRC (Proposed Model 2) technique is 0.9585, which is better than all experimented traditional

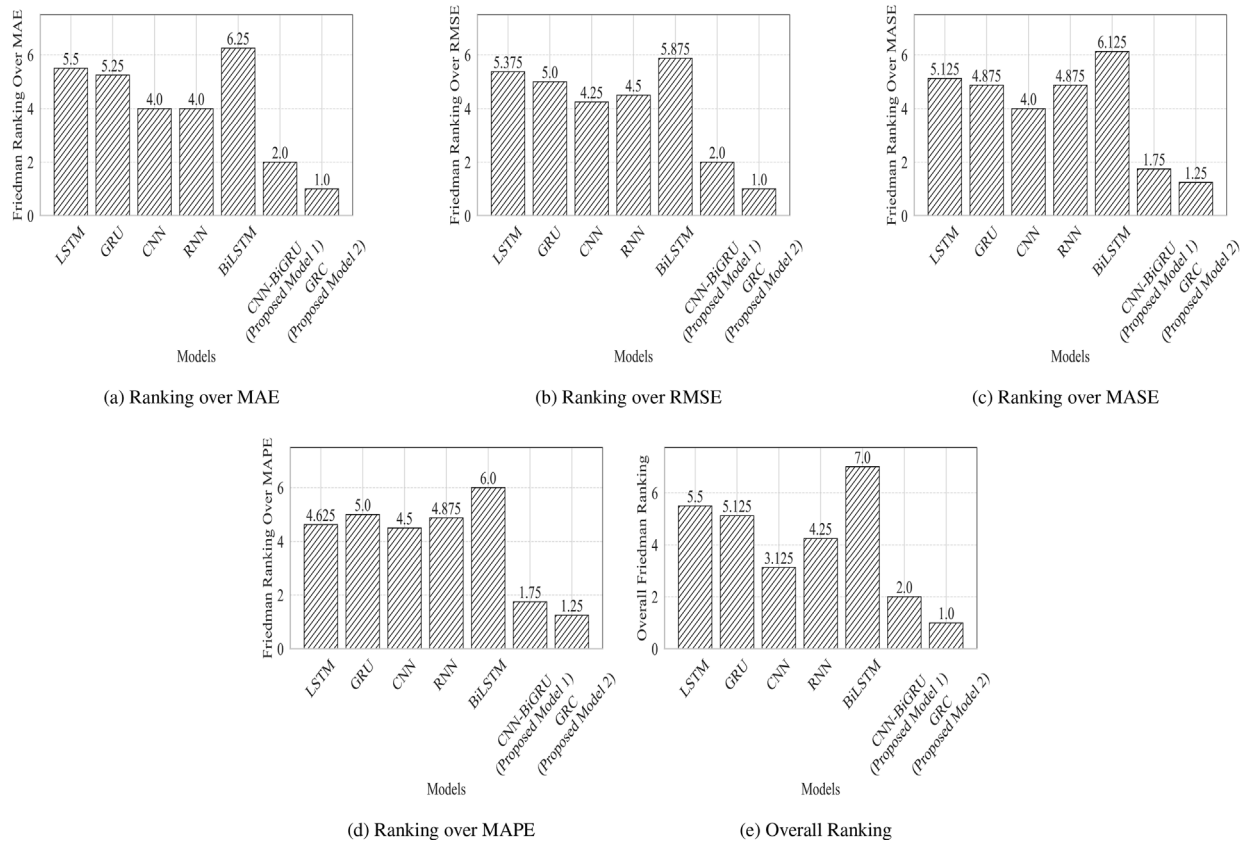


Fig. 17. Friedman Ranking of the models over different measures on all four variants of SSN data, where lower rank refers to the better model.

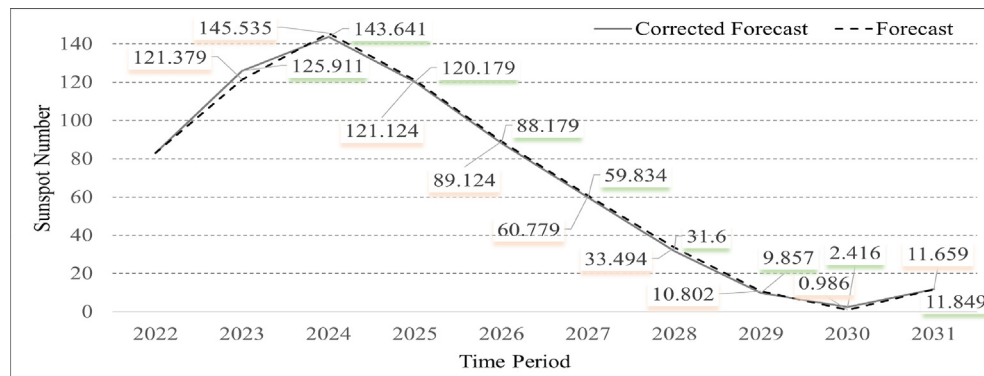


Fig. 18. Forecast of yearly average SSN for SC25 using proposed hybrid CNN-BiGRU (Proposed 1) along with the corrected predictions using GRC (Proposed 2) technique.

DL models. The R^2 value of the proposed model, which is 0.9510, is also improved by the GRC technique to 0.9585, which is represented in Fig. 11f. Fig. 12 illustrates the comparative view of different models with the proposed framework over yearly average SSN, where it can be observed that the R^2 value of the results obtained from the proposed framework is 0.7016, which is an improvement over the R^2 value of CNN-BiGRU model (Proposed Model 1), which is having an R^2 of 0.6282. The results obtained from the GRC technique are better than those of all other DL models that

are included here for comparative analysis. Fig. 13 represents the graph of actual vs predicted different traditional and proposed models over the test dataset of monthly average SSN data. In this plot, the trough value is the point at which the SSN reaches its minimum, while the crest is the point when the SSN reaches its maximum. Almost all models follow the pattern of test data, but the trough value is not mapped by most of the traditional models. The prediction line of the proposed model approximately coincides with actual test data, which depicts that it is capable of

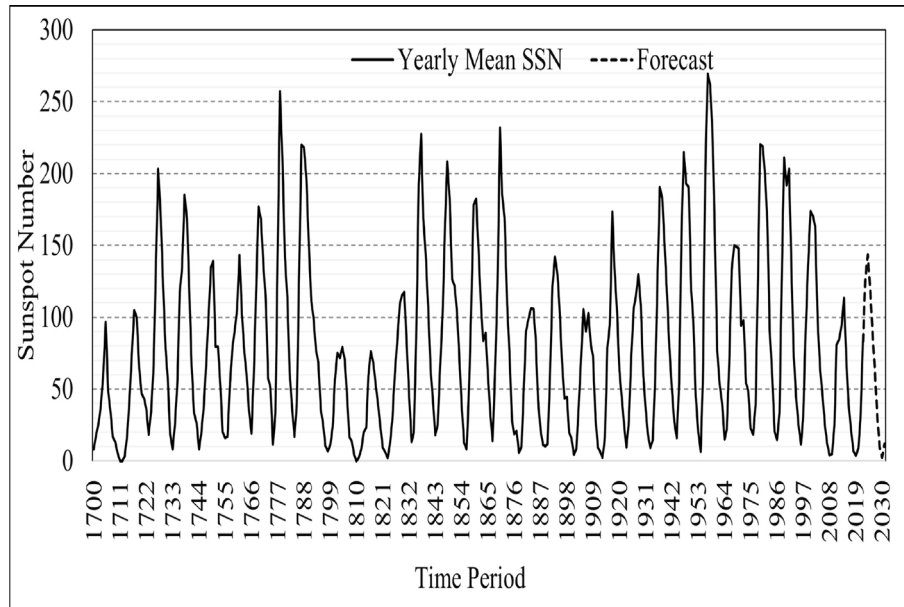


Fig. 19. Final forecast for the SC25 based on yearly averaged SSN utilizing GRC (Proposed 2) technique.

Table 4

Comparative analysis of performance metrics of different models over the four datasets with state-of-the-art work.

Data	Reference	Approach	RMSE	MAE	MASE	MSE	MAPE
Daily SSN	Peguerio and Carrasco (2023)	LSTM	40.1	-	-	-	-
	This article	CNN-BiGRU (Proposed Model 1)	11.88	8.83	0.6	141.13	1.38
	This article	GRC (Proposed Model 2)	10.86	8.54	0.65	117.94	1.36
	Nghiem et al. (2022)	CNN-LSTM	26.10	18.74	-	-	-
	Dang et al. (2022)	XgBoost + DL	25.70	19.82	-	-	-
Monthly Mean SSN	Pala and Atici (2019)	6-part LSTM	35.9	-	-	-	-
	Panigrahi et al. (2021)	PHM-MAX-ARIMA-ETS-SVM	22.73	16.549	0.91	-	-
	Hasoon and AL-Hashimi (2022)	DNN-LSTM	-	13.74	-	413.62	-
	Lee (2020)	EMD + LSTM	-	-	-	-	-
	This article	CNN-BiGRU (Proposed Model 1)	18.84	12.92	0.66	354.95	3.32
13-month Smoothed SSN	This article	GRC (Proposed Model 2)	8.16	5.92	0.31	66.59	1.73
	Ramadevi and Bingi (2022)	LSTM	-	-	-	10.85	-
	Ramadevi and Bingi (2022)	NAR NN	-	-	-	3.48	-
	Arfianti et al. (2021)	GRU	-	-	-	-	0.11
	Arfianti et al. (2021)	LSTM	-	-	-	-	0.12
Yearly SSN	This article	CNN-BiGRU (Proposed Model 1)	2.03	1.50	0.51	4.12	0.07
	This article	GRC (Proposed Model 2)	1.64	1.31	0.45	2.69	0.06
	Panigrahi et al. (2021)	PHM-MAX-ARIMA-ETS-SVM	25.46	18.99	0.61	-	-
	Elgamal (2020)	Stacked LSTM	23.478	-	-	-	-
	Kumar et al. (2023)	β SARMA + LSTM	-	-	0.46	-	-
Yearly SSN	This article	CNN-BiGRU (Proposed Model 1)	17.70	13.94	0.47	313.29	0.80
	This article	GRC(Proposed Model 2)	16.46	12.99	0.44	270.93	0.81

mapping up with the trough values of SSN during the minimum of the solar cycle. The line plot of the actual vs predicted value of SSN for the 13-month smoothed SSN is

depicted in Fig. 14. Similarly, the trough value is not effectively traced up by the RNN, GRU, BiLSTM and LSTM during the years 1997, 2009 and 2019. The CNN, CNN-

Table 5

Comparative analysis of predicted peak and corresponding time of different state-of-the-artwork in comparison to the proposed methods.

References	Approach	Forecasted Peak	Associated Time
Pala and Atici (2019)	LSTM	137.8±31.3	October, 2024
Prasad et al. (2022)	LSTM	171 ± 3.4	August 2023 ± 0.16
Pesnell and Schatten (2018)	Precursor Technique	135 ± 25	2025.2 ± 1.5
Benson et al. (2020)	DNN	106 ± 19.75	May, 2025 ± 1
Du (2020)	Statistical Forecasting	167.3	2023.2 ± 1.1
Upton and Hathaway (2018)	Dynamo-based Forecasting	110	-
Labonville et al. (2019)	Dynamo-based Forecasting	89± ²⁴ ₁₄	2025± ^{0.89} _{1.05}
Li et al. (2018)	Statistical Forecasting Technique	168.5 ± 16.3	October 2024
Kakad et al. (2017)	Shannon Entropy Technique	63 ± 11.3	-
Han and Yin (2019)	Statistical Forecasting	228.8 ± 40.5	2023
Zhu et al. (2023)	LSTM	213	January 2025
Zhu et al. (2022)	LSTM	143.6 ± 8.7	July 2025 ± 0.16
Okoh et al. (2018)	Hybrid Regression-NN	122.1 ± 18.2	January 2025 ± 0.5
Du (2022)	Correlation Statistics	135.5 ± 33.2	December 2024 ± 0.5
McIntosh et al. (2020)	Hilbert Transform	229 ± 76	-
Du (2020)	Waldmeier effect	-	2024.1 ± 0.8
Singh and Bhargawa (2017)	Hurst Exponent	102.8 ± 24.6	June 2024
Gonçalves et al. (2020)	Gaussian Process Regression	117	2024
Wang et al. (2021)	LSTM	114.3	2023
Covas et al. (2019)	Neural Network	57 ± 17	2022–2023
Courtillot et al. (2021)	Spectral Technique	97.6 ± 7.8	2026.2 ± 1
Sello (2019)	Non-linear prediction technique	107± 10	July 2023 ± 1
Xiong et al. (2021)	Precursor Technique	140.2	March 2024 ± 1
Guo et al. (2021)	Dynamo-based Forecasting	126	-
Petrovay et al. (2018)	Precursor Technique	130	2024
Kumar and Kumar (2023)	Stacked ConvLSTM1D	140.84	2024
This article	CNN-BiGRU (Proposed Model 1)	145.535	2024
This article	GRC (Proposed Model 2)	143.641	2024

BiGRU (Proposed Model 1), and its improved version utilizing the GRC (Proposed Model 2) technique is depicted in Fig. 14c, 14f, and 14g respectively shows that the trough value of the cycle is traced correctly. The crest value is not properly mapped by the CNN, while it is effectively mapped by the proposed model and its improved version. Similarly, the line plot of actual vs predicted yearly average of SSN test data is depicted in Fig. 15, which shows a lot of deviation while capturing the crest and trough of the two solar cycles present over there while performing predictions utilizing traditional DL models. The hybrid CNN-BiGRU model has improved the prediction in comparison to the other models, but the proposed framework contributed significantly in improving the prediction results and capability to catch the pattern of trough and crest of the solar cycles as illustrated in Fig. 15g. Similar improvement can be observed in Fig. 16g where the GRC (Proposed 2) technique is performing effectively on the daily SSN data. Fig. 16 represents the line plot of actual vs predicted daily SSN using the different DL approaches and the proposed framework.

Table 3 represents the mean of all performance measures along with their percentage of improvement with respect to the GRC (Proposed Model 2) technique for all experimented models over all four variants of the dataset. Table 3 shows that the performance is improved by 28.03 % based on RMSE while using the GRC (Proposed Model 2) technique with respect to the performance of LSTM, where RMSE is 22.87 for Yearly Mean SSN data. Similar

improvement can be observed for the other performance metrics and over other variants of data. The maximum improvement is observed for the MAPE over 13-month smooth SSN data in comparison to the BiLSTM model, where the improvement is 85.71%. A significant amount of improvement is also observed in comparison to the hybrid CNN-BiLSTM model (Proposed Model 1), where improvement in MASE ranges from 2.98% for daily SSN to 72.05% for the 13-month smooth SSN data. For the validation of the effectiveness of the proposed model and the proposed framework, a non-parametric statistical test is carried out using the Friedman ranking technique. The Friedman ranking is obtained over the four experimented datasets of SSN for the four performance measures individually. After obtaining the results, the overall Friedman ranking is obtained utilizing the results obtained in the previous step. In all the individual rankings along with the overall ranking, the GRC (Proposed Model 2) technique performed better, with the hybrid CNN-BiGRU (Proposed Model 1) model as the second-best ranking. The individual ranking, along with the overall Friedman ranking, has been depicted in Fig. 17.

The GRC (Proposed Model 2) method is utilized to forecast the remaining period of ongoing SC25 using an iterated one-step ahead forecast technique, and it has been observed that the SC25 is slightly stronger than SC24 and is expected to end in the year 2030. The forecast based on annual average SSN data is depicted in Fig. 19. The total time span of the ongoing cycle is also anticipated to be

11 years, and the peak is expected to be reached in the year 2024 with a peak value of 143.641. The forecast of ongoing SC25 using CNN-BiGRU(Proposed Model 1) model in comparison to GRC (Proposed Model 2) is depicted in Fig. 18 where the peak value obtained in both the forecast is observed for the year 2024. The forecast made by the CNN-BiGRU (Proposed Model 1) model suggests a peak with a value of 145.535, while the corrected forecast using the GRC (Proposed Model 2) technique suggests a peak with a value of 143.641. Similarly, the solar minimum is expected to be reached by the year 2030, according to both the proposed methods. However, there is a difference in the value of SSN for the solar minimum. CNN-BiGRU model suggests the solar minimum with a value of SSN as 0.986, while the corrected forecast using the GRC (Proposed Model 2) technique solar minimum is expected to have a value of SSN, which is equal to 2.416. The Table 5 illustrates the comparison of the peak of ongoing SC25 with respect to the other works existing in the literature with different methods and techniques, including precursor techniques, dynamo-based forecasting, Neural Network and statistical forecasting approach. Most researchers have observed that the peak of the SC25 will be achieved in the years 2024–2025. Based on our forecasting, we agree with the anticipation that the peak of the SC25 will be achieved in the year 2024.

The comparative analysis of the effectiveness of both the proposed in comparison to the different models from the literature are illustrated in Table 4. From Table 4, it can be observed that the proposed methods are performing better than other literature models based on different evaluation parameters over all four variants of the dataset.

5. Conclusion

In this study, the SSN dataset of four different variety is utilized to develop a generic model and framework that can predict the SSN more effectively. For this, a hybrid DL model composed of CNN and BiGRU is proposed. The model brought down the error measures significantly. A GRC technique is also proposed to improve the accuracy of the predictions. In this framework, the AdaBoost regressor is trained over the gradient and residual of training prediction and utilized to predict the residual for the forecasted data based on the gradient of predictions. Apart from these, the proposed model and its improved version utilizing the proposed framework are compared with traditional DL models, namely LSTM, GRU, CNN, BiLSTM, and RNN. The effectiveness of the proposed models is validated using Friedman's ranking of all the models obtained over different performance measures for all four kinds of data. A forecast of the ongoing solar cycle is also made utilizing the proposed model over the yearly average of SSN. The predictions are further enhanced by processing them through the proposed framework. The performance measures, peak and time span of SC25 are compared with the studies done on the SSN in the literature.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors acknowledge the contribution of the Royal Observatory of Belgium, SILSO World Data Center, for all four variants of sunspot data with different frequencies, which have been utilized to carry out the experiments in this research.

References

- Arfianti, U.I., Novitasari, D.C.R., Widodo, N., et al., 2021. Sunspot number prediction using gated recurrent unit (gru) algorithm. *IJCCS (Indonesian J. Comput. Cybernet. Syst.)* 15 (2), 141–152.
- Benson, B., Pan, W., Prasad, A., et al., 2020. Forecasting solar cycle 25 using deep neural networks. *Sol. Phys.* 295 (5), 65.
- Büyüksahin, Ü.Ç., Ertekin, Ş., 2019. Improving forecasting accuracy of time series data using a new arima-ann hybrid method and empirical mode decomposition. *Neurocomputing* 361, 151–163.
- Chattopadhyay, S., Jhajharia, D., Chattopadhyay, G., 2011. Trend estimation and univariate forecast of the sunspot numbers: development and comparison of arma, arima and autoregressive neural network models. *C.R. Geosci.* 343 (7), 433–442.
- Courtillot, V., Lopes, F., Le Mouél, J., 2021. On the prediction of solar cycles. *Sol. Phys.* 296, 1–23.
- Covas, E., Peixinho, N., Fernandes, J., 2019. Neural network forecast of the sunspot butterfly diagram. *Sol. Phys.* 294 (3), 24.
- Dang, Y., Chen, Z., Li, H., et al., 2022. A comparative study of non-deep learning, deep learning, and ensemble learning methods for sunspot number prediction. *Appl. Artif. Intell.* 36 (1), 2074129.
- Du, Z., 2020. The solar cycle: predicting the peak of solar cycle 25. *Astrophys. Space Sci.* 365 (6), 104.
- Du, Z., 2022. Predicting the maximum amplitude of solar cycle 25 using the early value of the rising phase. *Sol. Phys.* 297 (5), 61.
- Elgamal, M., 2020. Sunspot time series forecasting using deep learning. *Int. J. Comput. Inform. Technol.* (2279–0764) 9 (2).
- Gonçalves, Í.G., Echer, E., Frigo, E., 2020. Sunspot cycle prediction using warped gaussian process regression. *Adv. Space Res.* 65 (1), 677–683.
- Guo, W., Jiang, J., Wang, J.-X., 2021. A dynamo-based prediction of solar cycle 25. *Sol. Phys.* 296, 1–13.
- Han, Y., Yin, Z., 2019. A decline phase modeling for the prediction of solar cycle 25. *Sol. Phys.* 294 (8), 107.
- Hasoon, S.O., AL-Hashimi, M.M., 2022. Hybrid deep neural network and long short term memory network for predicting of sunspot time series. *Computer Science* 17 (3), 955–967.
- Kakad, B., Kakad, A., Ramesh, D.S., 2017. Shannon entropy-based prediction of solar cycle 25. *Sol. Phys.* 292, 1–12.
- Kumar, A., Kumar, V., 2023. Stacked 1d convolutional lstm (sconvlstm1d) model for effective prediction of sunspot time series. *Sol. Phys.* 298 (10), 1–33.
- Kumar, B., Yadav, N. et al. (2023). A novel hybrid model combining β sarma and lstm for time series forecasting. *Applied Soft Computing*, (p. 110019).
- Labonville, F., Charbonneau, P., Lemerle, A., 2019. A dynamo-based forecast of solar cycle 25. *Sol. Phys.* 294 (6), 82.
- Lee, T., 2020. Emd and lstm hybrid deep learning model for predicting sunspot number time series with a cyclic pattern. *Sol. Phys.* 295 (6), 82.
- Li, F., Kong, D., Xie, J., et al., 2018. Solar cycle characteristics and their application in the prediction of cycle 25. *J. Atmos. Solar Terr. Phys.* 181, 110–115.

- Marques, C., Leal-Júnior, A., Kumar, S., 2023. Multifunctional integration of optical fibers and nanomaterials for aircraft systems. *Materials* 16 (4), 1433.
- McIntosh, S.W., Chapman, S., Leamon, R.J., et al., 2020. Overlapping magnetic activity cycles and the sunspot number: forecasting sunspot cycle 25 amplitude. *Sol. Phys.* 295 (12), 1–14.
- Nghiem, T.-L., Le, V.-D., Le, T.-L., et al., 2022. Applying bayesian inference in a hybrid cnn-lstm model for time-series prediction. In: In 2022 International Conference on Multimedia Analysis and Pattern Recognition (MAPR). IEEE, pp. 1–6.
- Okoh, D., Seemala, G., Rabi, A., et al., 2018. A hybrid regression-neural network (hr-nn) method for forecasting the solar activity. *Space Weather* 16 (9), 1424–1436.
- Pala, Z., Atici, R., 2019. Forecasting sunspot time series using deep learning methods. *Sol. Phys.* 294 (5), 50.
- Panigrahi, S., Pattanayak, R.M., Sethy, P.K., et al., 2021. Forecasting of sunspot time series using a hybridization of arima, ets and svm methods. *Sol. Phys.* 296, 1–19.
- Peguero, J., Carrasco, V., 2023. A critical comment on "can solar cycle 25 be a new dalton minimum?". *Sol. Phys.* 298 (3), 48.
- Pesnell, W.D., Schatten, K.H., 2018. An early prediction of the amplitude of solar cycle 25. *Sol. Phys.* 293 (7), 112.
- Petrovay, K., Nagy, M., Gerjék, T., et al., 2018. Precursors of an upcoming solar cycle at high latitudes from coronal green line data. *J. Atmos. Solar Terr. Phys.* 176, 15–20.
- Prasad, A., Roy, S., Sarkar, A., et al., 2022. Prediction of solar cycle 25 using deep learning based long short-term memory forecasting technique. *Adv. Space Res.* 69 (1), 798–813.
- Prasad, A., Roy, S., Sarkar, A., et al., 2023. An improved prediction of solar cycle 25 using deep learning based neural network. *Sol. Phys.* 298 (3), 50.
- Ramadevi, B., Bingi, K., 2022. Time series forecasting model for sunspot number. In: In 2022 International Conference on Intelligent Controller and Computing for Smart Power (ICICCSPP). IEEE, pp. 1–6.
- Sello, S. (2019). Solar cycle activity: An early prediction for cycle# 25. arXiv preprint arXiv:1902.05294.
- SILSO World Data Center (). The International Sunspot Number. International Sunspot Number Monthly Bulletin and online catalogue.
- Singh, A., Bhargawa, A., 2017. An early prediction of 25th solar cycle using hurst exponent. *Astrophys. Space Sci.* 362 (11), 199.
- Upton, L.A., Hathaway, D.H., 2018. An updated solar cycle 25 prediction with aft: The modern minimum. *Geophys. Res. Lett.* 45 (16), 8091–8095.
- Vokhmyanin, M., Arlt, R., Zolotova, N., 2020. Sunspot positions and areas from observations by thomas harriot. *Sol. Phys.* 295 (3), 39.
- Wang, Q.-J., Li, J.-C., Guo, L.-Q., 2021. Solar cycle prediction using a long short-term memory deep learning model. *Res. Astron. Astrophys.* 21 (1), 012.
- Xiong, Y., Lu, J., Zhao, K., et al., 2021. Forecasting solar cycle 25 using comprehensive precursor combination and multiple regression technique. *Mon. Not. R. Astron. Soc.* 505 (1), 1046–1052.
- Zhu, H., Chen, H., Zhu, W., et al., 2023. Predicting solar cycle 25 using an optimized long short-term memory model based on sunspot area data. *Adv. Space Res.* 71 (8), 3521–3531.
- Zhu, H., Zhu, W., He, M., 2022. Solar cycle 25 prediction using an optimized long short-term memory mode with fl0. 7. *Sol. Phys.* 297 (12), 157.